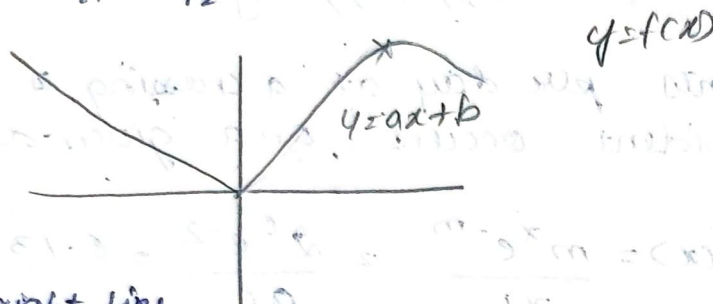


Module - 3

Curve Fitting

consider the given data (observed data).

$x: x_1, x_2, \dots, x_n$ to fit a curve $y=f(x)$
 $y: y_1, y_2, \dots, y_n$



I straight line

In the above diagram we can fit a straight line in the form

$$y = ax + b \text{ or } y = a + bx$$

II parabola

To fit a 2nd degree parabola

$$y = ax^2 + bx + c \text{ or } y = a + bx + cx^2$$

III To fit an exponential curve

$$y = ae^{bx} \text{ or } y = ax^b$$

in this method we have to take log on both sides & converted into type I

Normal equation

$$y = ax + b \text{ — ①}$$

$$\sum y = a \sum x + b \sum 1 = a \sum x + bn \text{ — ②}$$

multiple 'x' in ①

$$\sum xy = a \sum x^2 + b \sum x \text{ — ③}$$

Solve ② & ③

We get a & b values

II $y = a + bx$

normal eqⁿ

$$\sum y = a \sum 1 + b \sum x \text{ — ①}$$

$$\sum yx = a \sum x + b \sum x^2 \text{ — ②}$$

III $y = ax^2 + bx + c$ — ①

normal eqⁿ

$$\sum y = a \sum x^2 + b \sum x + c \sum 1$$

$$\sum xy = a \sum x^3 + b \sum x^2 + c \sum x$$

multiply x in ①

multiple x^2 in ①
 $\sum x^2 y = a \sum x^4 + b \sum x^3 + c \sum x^2$

$$y = a e^{bx}$$

take log on b.s

$$\log y = \log a + \log e^{bx}$$

$$\log y = \log a + bx \log e$$

$$\log y = \log a + bx \quad (1)$$

$$y = A + bx \quad \text{--- ①}$$

$$\text{when } y = \log y \quad A = \log a$$

Normal eqn

$$\sum y = A \sum n + b \sum x$$

$$\sum xy = A \sum x + b \sum x^2$$

$$y = ax^b$$

$$\log y = \log a + b \log x$$

$$\log y = \log a + b \log x$$

$$y = A + bX$$

Normal eqn

$$\sum y = A \sum n + b \sum X$$

multiply 'X'

$$\sum XY = A \sum X + b \sum X^2$$

1. Fit a straight line $y = ax + b$ for the following data by a method of least square

$$x: \quad 1 \quad 3 \quad 4 \quad 6 \quad 8 \quad 9 \quad 11 \quad 14$$

$$y: \quad 1 \quad 2 \quad 4 \quad 4 \quad 5 \quad 7 \quad 8 \quad 9$$

Given $y = ax + b$ --- ①

Normal eqn

$$\sum y = a \sum x + b \sum n \quad \text{--- ②}$$

multiply x for ①

$$\sum xy = a \sum x^2 + b \sum x \quad \text{--- ③}$$

$$n=8$$

x	y	xy	x ²
1	1	1	1
3	2	6	9
4	4	16	16
6	4	24	36
8	5	40	64
9	7	63	81
11	8	88	121
14	9	126	196
Σx	Σy	Σxy	Σx^2
= 56	= 40	= 364	= 524

from ① & ②

$$40 = a(56) + b(8) \quad \text{--- ④}$$

$$364 = a(324) + b(56) \quad \text{--- ⑤}$$

Solve ④ & ⑤

$$a = 0.6363, \quad b = 0.5454$$

$$y = (0.6363)x + (0.5454)$$

2. p is pull required to lift a load w by means of pulley block. find the linear law of the form $p = mw + c$ connecting p & w, using the following data where p & w are taken in kg-weight kg-wt. compute p when w = 150 kgwt

P:	12	15	21	25
w:	50	70	100	120

$$\rightarrow p = mw + c \quad \text{--- ①}$$

Normal eqn.

$$\Sigma p = m \Sigma w + cn \quad \text{--- ②}$$

multiply w on b.s

$$\Sigma pw = m \Sigma w^2 + c \Sigma w \quad \text{--- ③}$$

$$n=4$$

P	w	Pw	w ²
12	50	600	2500
15	70	1050	4900
21	100	2100	10000
25	120	3000	14400
$\Sigma p = 73$	$\Sigma w = 340$	$\Sigma pw = 6750$	$\Sigma w^2 = 31800$

from ② & ③

$$73 = m(340) + c(4) \quad \text{--- ④}$$

$$6750 = m(3180) + c(340) \quad \text{--- ⑤}$$

$$m = 0.1879 \quad c = 2.275$$

from ①

$$p = mw + c$$

$$p = 0.1879w + 2.275$$

$$w = 150 \text{ kg-wt}$$

$$p = 0.1879(150) + 2.275 = \underline{30.46}$$

3. A data science team is investigating how the size of the training data set influence the prediction accuracy of the machine learning model the following observation were recorded during the experimentation. assuming a linear relation b/w the data set size & accuracy determine the eqn. of best fit line $y = ax + b$ using the method of least square.

x:	1	2	3	4	6	8
y:	2.4	3.0	3.6	4.0	5.0	6.0

→ given $y = ax + b$ --- ①

normal eqn.,

$$\sum y = a \sum x + b \sum 1 \quad \text{--- ②}$$

multiply x on b.s

$$\sum yx = a \sum x^2 + b \sum x \quad \text{--- ③}$$

n = 6

x	y	yx	x ²
1	2.4	2.4	1
2	3.0	6	4
3	3.6	10.8	9
4	4.0	16	16
6	5.0	30	36
8	6.0	48	64

$$\begin{array}{llll} \sum x = & \sum y = & \sum yx = & \sum x^2 = \\ 24 & 24 & 113.2 & 130 \end{array}$$

from ② & ③

$$24 = a(24) + b(6) \quad \text{--- ④}$$

$$113.2 = a(130) + b(24) \quad \text{--- ⑤}$$

solve ④ & ⑤

$$a = 0.505 \quad b = 1.976$$

from ① $y = ax + b$

$$y = (0.505)x + 1.976$$

4. The manufacturing company producing industrial grade steel components as recorded in its annual production (in tons) at 10th year interval as follows.

Year :	1961	1971	1981	1991	2001
production : - in tons	8	10	12	10	16

assuming that the produ trend follows a linear pattern
fit a straing line to the data using the method of
least square & estimate the expected prod. for the
year 2006

→ given $y = ax + b$ — ①

Normal eqn

$\sum y = a \sum x + b n$ — ②

multiply x on b.s

$\sum xy = a \sum x^2 + b \sum x$ — ③

x	y	xy	x ²
1961	8	15688	3845521
1971	10	19710	3884841
1981	12	23772	3924361
1991	10	19910	3964081
2001	16	32016	4004001

$\sum x^2$ 9905	$\sum y$ 56	$\sum xy$ 111096	$\sum x^2$ 19622805
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from ② & ③
 $56 = a(9905) + b(5)$

$111096 = a(19622805) + b(9905)$

$a = 0.6$ $b = -305.76$

$y = ax + b$

$y = 0.6x - 305.76$

at $x = 2006$

$y = 0.6(2006) - 305.76$

$y = 15.2 =$

5. Fit a second degree parabola $y = a + bx + cx^2$ for the following data

x:	2	4	6	8	10
y:	3.07	12.85	31.47	57.38	91.29

assume the relation b/w signal cycle length & average delay can be modelled.

by quadratic eqn,

$$y = a + bx + cx^2 \quad \text{--- (1)}$$

Normal eqn

$$\sum y = an + b \sum x + c \sum x^2 \quad \text{--- (2)}$$

multiply 'x' on b.s in (1)

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3 \quad \text{--- (3)}$$

multiple x^2 on b.s in (1)

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4 \quad \text{--- (4)}$$

x	y	xy	$x^2 y$	x^2	x^3	x^4
2	3.07	6.14	12.28	4	8	16
4	12.85	51.4	205.6	16	64	256
6	31.47	188.82	1132.92	36	216	1296
8	57.38	459.04	3672.32	64	512	4096
10	91.29	912.9	9129	100	1000	10000
$\sum x$	$\sum y$	$\sum xy$	$\sum x^2 y$	$\sum x^2$	$\sum x^3$	$\sum x^4$
30	196.06	1618.3	14152.12	220	1800	15664

from (2) (3) & (4)

$$196.06 = a(5) + b(30) + c(220)$$

$$1618.3 = a(30) + b(220) + c(1800)$$

$$14152.12 = a(220) + b(1800) + c(15664)$$

$$a = 0.696, \quad b = -0.855, \quad c = 0.991$$

from (1)

$$y = 0.696 - 0.855x + 0.991x^2$$

6. Fit a second degree parabola $y = ax^2 + bx + c$ in the least square sense for the following data & hence estimate y at $x=6$

x:	1	2	3	4	5
y:	10	12	13	16	19

$$y = ax^2 + bx + c \quad \text{--- ①}$$

Normal eqn x

$$\sum y = a \sum x^2 + b \sum x + cn \quad \text{--- ②}$$

multiply x on ①

$$\sum xy = a \sum x^3 + b \sum x^2 + c \sum x \quad \text{--- ③}$$

multiply x^2 in ①

$$\sum x^2 y = a \sum x^4 + b \sum x^3 + c \sum x^2 \quad \text{--- ④}$$

Ans

x	y	xy	x^2y	x^2	x^3	x^4
1	10	10	10	1	1	1
2	12	24	48	4	8	16
3	13	39	117	9	27	81
4	16	64	256	16	64	256
5	19	95	475	25	125	625
$\sum x = 15$	$\sum y = 70$	$\sum xy = 232$	$\sum x^2y = 906$	$\sum x^2 = 55$	$\sum x^3 = 225$	$\sum x^4 = 979$

$$70 = a(55) + b(15) + c(5)$$

$$232 = a(225) + b(55) + c(15)$$

$$906 = a(979) + b(225) + c(55)$$

$$a = 0.285, \quad b = 0.485, \quad c = 9.4$$

$$y = ax^2 + bx + c$$

$$y = 0.285x^2 + 0.485x + 9.4$$

$$\text{at } x = 6$$

$$y = 22.57$$

Q.29

7. Relation b/w the size of HP data x in gb & corresponding processing time y in sec. & following data using quadratic model $y = a + bx + cx^2$ use the method of least square method determine the best

x	-2	-1	0	1	2
y	-3.150	-1.390	0.620	2.880	5.378

$y = a + bx + cx^2$ — ①
 normal eqn

$\Sigma y = an + b \Sigma x + c \Sigma x^2$ — ②

multiply x on — ①

$\Sigma xy = a \Sigma x + b \Sigma x^2 + c \Sigma x^3$ — ③

multiply x^2 on ①

$\Sigma x^2 y = a \Sigma x^2 + b \Sigma x^3 + c \Sigma x^4$ — ④

x	y	xy	$x^2 y$	x^2	x^3	x^4
-3	1.50	-4.50	13.50	9	-27	81
-2	1.390	-2.78	5.56	4	-8	16
-1	0.620	-0.62	0.62	1	-1	1
0	2.880	0	0	0	0	0
1	2.880	2.88	2.88	1	1	1
2	5.378	10.756	21.512	4	8	16
$\Sigma x = 0$	$\Sigma y = 4.338$	$\Sigma xy = 21.326$	$\Sigma x^2 y = 38.602$	$\Sigma x^2 = 10$	$\Sigma x^3 = 0$	$\Sigma x^4 = 34$

$4.338 = a(5) + b(0) + c(10)$ — ①

$21.326 = a(0) + b(10) + c(0)$ — ②

$38.602 = a(10) + b(0) + c(34)$ — ③

$a = 0.621, b = 2.132, c = 0.123$

$y = a + bx + cx^2$

$y = 0.621 + 2.132x + 0.123x^2$

$y = a x^b$ (or) $a \cdot e^{bx}$ (type ②)

in this method we have to take log on b.s

1. Fit a curve $y = ax^b$ for the following data

x :	1	2	3	4
y :	1.7	1.8	2.3	3.2

$y = ax^b$

take log on b.s

$$\log y = \log a x^b$$

$$\log y = \log a + \log x^b$$

$$\log y = \log a + b \log x$$

$$y = A x^b \quad \text{--- (1)}$$

is straight line

where $y = \log y$

$$A = \log a$$

$$X = \log x$$

$$\log x = \log x$$

x	y	$X = \log x$	$Y = \log y$	X^2	XY
1	1.7	0	0.2304	0	0
2	1.8	0.3010	0.2582	0.090601	0.0776
3	2.3	0.4771	0.3617	0.2276	0.1727
4	3.2	0.6020	0.5051	0.3624	0.3032
$\Sigma x = 10$	$\Sigma y = 9$	$\Sigma X = 1.3801$	$\Sigma Y = 1.3554$	$\Sigma X^2 = 30$	$\Sigma XY = 25$

$$\Sigma y = An + b \Sigma x$$

$$\Sigma xy = A \Sigma x + b \Sigma x^2$$

$$a =$$

x	y	$X = \log x$	$Y = \log y$	X^2	XY
1	1.7	0	0.530	0	0
2	1.8	0.693	0.587	0.480	0.4067
3	2.3	1.098	0.832	1.205	0.9135
4	3.2	1.386	1.163	1.920	1.6119
		3.177	3.112	3.605	2.9321

$$3.112 = A(4) + b(3.177)$$

$$2.932 = A(3.177) + b(3.605)$$

$$A = 0.440$$

$$b = 0.425$$

$$\hookrightarrow \log a = 0.440$$

$$a = e^{0.440}$$

$$a = 1.5527$$

from (1)

$$y = ax^b$$

$$y = 1.5527x^{0.425}$$

2. fit a curve $y = ae^{bx}$ for the following data

x	1	2	3	4	5
y	10	12	13	16	19

$$y = ae^{bx}$$

$$\log y = \log ae^{bx}$$

$$\log y = \log a + \log e^{bx}$$

$$\log y = \log a + bx \log e \quad \log e = 1$$

$$\log y = \log a + bx \quad \text{--- (1)}$$

$y = A + bx$ --- (1) is straight line

$$\Sigma y = An + b \Sigma x \quad \text{--- (2)}$$

multiply 'x' in eqn (2)

$$\Sigma yx = A \Sigma x + b \Sigma x^2 \quad \text{--- (3)}$$

x	y	$u = \log y$	x^2	yx
1	10	2.3025	1	2.3025
2	12	2.4849	4	4.9698
3	13	2.5649	9	7.6947
4	16	2.7725	16	41.09
5	19	2.9444	25	14.722

$$\Sigma x = 15 \quad \Sigma y = 70 \quad \Sigma u = 13.0692 \quad \Sigma x^2 = 55 \quad \Sigma yx = 40.779$$

$$13.0692 = A(5) + b(15)$$

$$40.779 = A(15) + b(55)$$

$$A = 2.1424$$

$$b = 0.1571$$

$$\log a = 2.1424$$

$$a = e^{2.1424}$$

$$a = 8.5198$$

$$y = ae^{bx}$$

$$y = 8.5198 e^{0.1571x}$$

$$13.0692 = A(5) + b(15)$$

$$40.779 = A(15) + b(55)$$

$$A = 6.3637 \quad b = 0.1571$$

Co-relations & co-efficient of co-relations

Co-variation of 2 independent magnitude is known as co-relations. If 2 variable x & y are related in such a way that increasing or decreasing in one of them corresponding to increasing or decreasing in other way.

ex: Marks of individual in 2 subjects.

Ratio height & weight of 2 individuals

Karl Pearson's

Co-efficient of co-relation

$$r = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{\sigma_{xy}}{\sqrt{\sigma_x^2 \sigma_y^2}}$$

$$\text{where } \sigma_x^2 = \frac{\sum x^2}{n} - (\bar{x})^2$$

$$\sigma_y^2 = \frac{\sum y^2}{n} - (\bar{y})^2$$

Line of Regression y on x

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$z = \frac{x - \bar{x}}{\sigma_x}$$

$$\sigma_{x-y}^2 = \sigma_z^2 = \frac{\sum z^2}{n} - \left(\frac{\sum z}{n}\right)^2$$

Line of regression x on y

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$r = \sqrt{\text{coeff of } x \cdot \text{coeff of } y}$$

1. Obtain the lines of regression & hence find co-efficient of co-relations for the following data

x :	1	2	3	4	5	6	7	8
y :	9	8	10	12	11	13	14	15

$$\bar{x} = \frac{\sum x}{n} = \frac{28}{7} = 4$$

$$\bar{y} = \frac{\sum y}{n} = \frac{77}{7} = 11$$

x	y	x^2
1	9	1
2	8	4
3	10	9
4	12	16
5	11	25
6	13	36
7	14	49

y^2	$z = x - \bar{x}$	z^2
81	-8	64
64	-6	36
100	-7	49
144	-8	64
121	-6	36
169	-7	49
196	-7	49

Rank coordinates b/w Judge 1 & 2 $\rho_{12} = \frac{1 - 6 \sum d_{12}^2}{n(n^2 - 1)}$

$$= \frac{1 - 6(64)}{10(10^2 - 1)} = 0.6121$$

Judge 1 & 3

$$\rho_{13} = \frac{1 - 6 \sum d_{13}^2}{n(n^2 - 1)} = \frac{1 - 6(18)}{10(10^2 - 1)} = 0.8909$$

Judge 2 & 3

$$\rho_{23} = \frac{1 - 6 \sum d_{23}^2}{n(n^2 - 1)} = \frac{1 - 6(94)}{10(10^2 - 1)} = 0.4303$$

Among these highest is $\rho_{13} = 0.8909$

Hence judge 1 & 3 have common taste in respect of beauty

x:	1	3	4	2	5	8	9	10	13	15	70
y:	8	6	10	8	12	16	16	10	32	32	150

$$\bar{x} = \frac{\sum x}{n} = \frac{70}{10} = 7$$

$$\bar{y} = \frac{\sum y}{n} = \frac{150}{10} = 15$$

$$\bar{y} = \frac{\sum y}{n} = \frac{150}{10} = 15$$

x	y	x ²	y ²	z = x - y	z ²
1	8	1	64	-7	49
3	6	9	36	-3	9
4	10	16	100	-6	36
2	8	4	64	-6	36
5	12	25	144	-7	49
8	16	64	256	-8	64
9	16	81	256	-7	49
10	10	100	100	0	0
13	32	169	1024	-19	361
15	32	225	1024	-17	289

$$\sum x = 70 \quad \sum y = 150 \quad \sum x^2 = 694 \quad \sum y^2 = 3068 \quad \sum z = -80 \quad \sum z^2 = 942$$

where

$$\sigma_x^2 = \frac{\sum x^2 - (\sum x)^2}{n} = \frac{694 - (70)^2}{10} = \frac{694 - 4900}{10} = \frac{-4206}{10} = -420.6$$

$$\sigma_y^2 = \frac{\sum y^2 - (\sum y)^2}{n} = \frac{3068 - (150)^2}{10} = \frac{3068 - 22500}{10} = \frac{-19432}{10} = -1943.2$$

$$\sigma_z^2 = \frac{\sum z^2 - (\sum z)^2}{n} = \frac{942 - (-80)^2}{10} = \frac{942 - 6400}{10} = \frac{-5458}{10} = -545.8$$

$$r = \frac{\sigma_x^2 + \sigma_y^2 + \sigma_z^2}{2\sigma_x\sigma_y}$$

$$r = \frac{20.4 + 81.8 - 30.2}{2 \times 4.5 \times 9.04}$$

$$r = 0.881$$

x	y	x ²	y ²	z = x - y	z ²
1	10	1	100	-9	81
2	12	4	144	-10	100
3	16	9	256	-13	169
4	28	16	784	-24	576
5	25	25	625	-20	400
6	36	36	1296	-30	900
7	41	49	1681	-34	1156
8	49	64	2401	-41	1681
9	40	81	1600	-31	961
10	50	100	2500	-40	1600

$$\Sigma x = 55 \quad \Sigma y = 301 \quad \Sigma x^2 = 385 \quad \Sigma y^2 = 11387 \quad \Sigma z = -252 \quad \Sigma z^2 = 7624$$

Karl Pearson's Coeff of correlation

$$r = \frac{\sqrt{\Sigma x^2 + \Sigma y^2 - \Sigma (x-y)^2}}{2\sigma_x \sigma_y}$$

$$r = \frac{8.25 + 196.21 - 127.36}{2 \times 2.872 \times 14.007}$$

$$= 0.9582 \approx 0.96$$

Note $\Rightarrow r = \pm 1$ we say that x & y are perfect co-relations
 if $r = 0$ we say that x & y are non correlations

3. Obtain the lines of regressions & hence find co-efficient of correlation for the data

$$(x: 1 \quad 3 \quad 3 \quad 4 \quad 2 \quad 4)$$

(y: 8)

$$\sum x^2 = 28$$

$$\sum y = 77 \quad \sum x^2 = 140$$

$$\sum y^2 = 875$$

$$\sum z = -49 \quad \sum z^2 = 347$$

where

$$\sigma_x^2 = \frac{\sum x^2}{n} - (\bar{x})^2 = \frac{140}{7} - (4)^2 = 4$$

$$\sigma_y^2 = \frac{\sum y^2}{n} - (\bar{y})^2 = \frac{875}{7} - (11)^2 = 4$$

$$\sigma_z^2 = \frac{\sum z^2}{n} - (\bar{z})^2 = \frac{347}{7} - (-7)^2 = 0.571$$

$$r = \frac{\sigma_x^2 + \sigma_y^2 + \sigma_z^2}{2\sigma_x \sigma_y}$$

$$\sigma_x = 2$$

$$\sigma_y = 2$$

$$r = \frac{4 + 4 - 0.571}{2(2)(2)}$$

$$r = 0.9286$$

regression lines are y on x

$$(y - \bar{y}) = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 11 = 0.93 \frac{(2)}{2} (x - 4)$$

$$y - 11 = 0.93 (x - 4)$$

$$y = 0.93x - 0.93 \times 4 + 11$$

$$y = 0.93x + 7.2856$$

regression lines x on y

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$x - 4 = 0.93 \frac{(2)}{2} (y - 11)$$

$$x - 4 = 0.93y - 11 \times 0.93$$

$$x = 0.93y - 11 \times 0.93 + 4$$

$$x = 0.93y - 6.214$$

or find Karl Pearson's co-efficient of correlation.

2. Find the co-efficient of correlation using the following data

x:	1	2	3	4	5	6	7	8	9	10
y:	10	12	16	28	25	36	41	49	40	50

→

$$\bar{x} = \frac{\sum x}{n} = \frac{55}{10} = 5.5$$

$$\bar{z} = \frac{\sum z}{n} = \frac{-252}{10}$$

$$\bar{y} = \frac{\sum y}{n} = \frac{307}{10} = 30.7$$

$$= -25.2$$

$$\sigma_x^2 = \frac{\sum x^2}{n} - (\bar{x})^2 \Rightarrow \frac{385}{10} - (5.5)^2 = 8.25 \quad \sigma_x = 2.872$$

$$\sigma_y^2 = \frac{\sum y^2}{n} - (\bar{y})^2 \Rightarrow \frac{11387}{10} - (30.7)^2 = 196.21 \quad \sigma_y = 14.007$$

$$\sigma_z^2 =$$

$$\frac{\sum z^2}{n} - (\bar{z})^2 \Rightarrow \frac{7624}{10} - (-25.2)^2 = 127.36$$

$$x = \frac{-34 \pm \frac{1}{2}}{2}$$

$$y = \frac{-1}{6}x + \frac{4}{6}$$

note: r lies b/w $-1 < r < 1$

$$r = \sqrt{\text{co of } x \cdot \text{co of } y}$$

$$= \sqrt{-\frac{1}{6} \times -3/2}$$

$$r = 0.5 \Rightarrow \boxed{r = -0.5}$$

x co of is -ve
 y co of is -ve
so r is -ve

find the means of x 's & y 's & the correlation coefficient.
find σ_y if $\sigma_x = 3$

the given lines are passing through

$$8x - 10y + 66 = 0 \quad \text{--- (1)}$$

$$40x - 18y = 214 \quad \text{--- (2)}$$

$$\bar{x} = 13, \quad \bar{y} = 17$$

$$y = \frac{8}{10}x + \frac{66}{10} \quad \text{--- (3)}$$

$$x = \frac{18}{40}y - \frac{214}{40} \quad \text{--- (4)}$$

$$r = \sqrt{\text{co of } x \cdot \text{co of } y}$$

$$= \sqrt{\frac{8}{10} \times \frac{18}{40}}$$

$$r = 0.6$$

from (1) & (2)

$$\frac{r \sigma_y}{\sigma_x} = \frac{8}{10}$$

$$\text{& } \frac{r \sigma_x}{\sigma_y} = \frac{18}{40}$$

$$0.6 \frac{\sigma_y}{3} = \frac{8}{10}$$

$$1.8 \cancel{4} 0.8$$

$$\sigma_y = \frac{8}{10} \times \frac{3}{0.6}$$

$$\sigma_y = 4$$

7. given mean

x -series y -series

mean 18 100

S.D 14 20

& $r = 0.8$, write down the eqn of lines of regressions & hence find most probable value of y when $x = 70$

$$(y - \bar{y}) = \frac{r \sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 100 = 0.8 \frac{20}{14} (x - 18)$$

$$y = 1.14x + 79.48 \quad \text{--- (1)}$$

$$(x - \bar{x}) = \frac{r \sigma_x}{\sigma_y} (y - \bar{y})$$

$$x - 18 = 0.8 \frac{14}{20} (y - 100)$$

$$x = 0.56y - 38 \quad \text{--- (2)}$$

put $x = 70$ in eqn (1)

$$y = 1.14(70) + 79.48$$

$$y = 159.28$$

lines of regression y on x

$$(y - \bar{y}) = r \frac{s_y}{s_x} (x - \bar{x})$$

$$y - 15 = 0.881 \frac{(9.04)}{(4.5)} (x - 7)$$

$$y = 1.76x + 2.65$$

line of regression x on y

$$(x - \bar{x}) = r \frac{s_x}{s_y} (y - \bar{y})$$

$$x - 7 = 0.881 \frac{(4.5)}{(9.04)} (y - 15)$$

$$x = 0.4395y + 0.4013$$

4. The given eqn. of regression lines

$x = 19.13 - 0.87y$, $y = 11.64 - 0.5x$, compute mean of x 's, mean of y 's, and co-efficient of correlations.

→ we have to compute $\bar{x}$, $\bar{y}$ & r .

the given regression lines passing through $\bar{x}$, $\bar{y}$ i.e.

$$\bar{x} = 19.13 - 0.87\bar{y}$$

$$\Rightarrow \bar{x} + 0.87\bar{y} = 19.13 \quad \text{--- ①}$$

$$\bar{y} = 11.64 - 0.5\bar{x}$$

$$0.5\bar{x} + \bar{y} = 11.64 \quad \text{--- ②}$$

Solve ① & ②

$$\bar{x} = 15.93, \quad \bar{y} = 3.67$$

co-eff of correlation $r = \sqrt{\text{co. of } x \cdot \text{co. of } y}$

$$= \sqrt{-0.5 \times -0.87}$$

$$r = -0.6595$$

5. Compute $\bar{x}$, $\bar{y}$ & r from the following eqn. of regression lines $2x + 3y + 1 = 0$ and $x + 6y - 4 = 0$

→ The regression lines are passing through $\bar{x}$, $\bar{y}$ is given by

$$2\bar{x} + 3\bar{y} = -1 \quad \text{--- ①}$$

$$\bar{x} + 6\bar{y} = 4 \quad \text{--- ②}$$

Solve ① & ②

$$\bar{x} = -2 \quad \bar{y} = 1$$

Estimated (x) : 300 450 800 250 500 975 475 400

Actual (y) : 273 486 734 297 681 872 396 457

x : 300 450 800 250 500 975 475 400

y : 273 486 734 297 681 872 396 457

R_x : 7 5 2 8 3 1 4 6

R_y : 8 4 2 7 3 1 6 5

$d = R_x - R_y$: -1 1 0 1 0 0 -2 1

d^2 : 1 1 0 1 0 0 4 1

$$\sum d^2 = 8 \quad n = 8$$

$$r = \frac{1 - 6 \sum d^2}{n(n^2 - 1)} = \frac{1 - 6(8)}{8(8^2 - 1)} = 0.9047$$

In a beauty contest 10 participants are ranked by 3 judges determine which pair of the judges as a common taste in respective of beauty.

Judge 1	4	7	2	1	5	10	9	6	3	8
Judge 2	6	5	1	2	9	10	3	7	4	8
Judge 3	6	8	2	1	5	9	10	3	4	7

Judge 1	4	7	2	1	5	10	9	6	3	8
Judge 2	6	5	1	2	9	10	3	7	4	8
Judge 3	6	8	2	1	5	9	10	3	4	7

d_{12} : -2 +2 1 -1 -4 0 +6 -1 -1 0

d_{13} : -2 -1 0 0 0 1 -1 3 -1 1

d_{23} : 0 -3 -1 1 4 1 -7 +4 0 1

d_{12}^2 : 4 4 1 1 16 0 36 1 1 0

d_{13}^2 : 4 1 0 0 0 1 1 9 1 1

d_{23}^2 : 0 9 1 1 16 1 49 16 0 1

2. Calculate the rank of the co-relation of the following data.

x:	68	64	75	50	69	80	76	40	55
y:	62	58	68	45	81	60	69	48	30
R _x :	5	6	3	8	4	1	2	9	7
R _y :	4	6	3	9	1	5	2	8	7
d = R _x - R _y :	1	0	0	-1	3	-4	0	1	0
d ² :	1	0	0	1	9	16	0	1	0

$$\sum d^2 = 28, n = 9$$

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} = 1 - \frac{6(28)}{9(9^2 - 1)} = 0.767$$

3. The rank of 10 students of same batch in 2 sub A & B are given below. Calculate the rank of the co-relation

rank of A:	1	2	3	4	5	6	7	8	9	10
rank of B:	6	7	5	10	3	9	4	1	8	2

R _x :	1	2	3	4	5	6	7	8	9	10
R _y :	6	7	5	10	3	9	4	1	8	2
d = R _x - R _y :	-5	-5	-2	-6	2	-3	3	7	1	8
d ² :	25	25	4	36	4	9	9	49	1	64

$$\sum d^2 = 226, n = 10$$

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} \Rightarrow 1 - \frac{6(226)}{10(10^2 - 1)} = 0.37$$

3. Random sample of recent repair jobs was selected & estimated cost & actually cost were recorded. Compute the rank and Spearman rank co-relation.

Spearman's coeff. of co-relation (Rank co-relation)

Spearman rank co-relation is measured for co-relation of 2 ranked variable. Spearman co-relation replace observation by their ranks in the calculation of co-relation.

assign rank to x series data & y series data highest value of rank is 1.

R_x denotes rank assign for x data

R_y denotes rank assign for y data.

let $d = R_x - R_y$ difference in a rank

x x_1 x_2 ... x_n

y y_1 y_2 ... y_n

R_x R_{x1} R_{x2} ... R_{xn}

R_y R_{y1} R_{y2} ... R_{yn}

$d = R_x - R_y$ $R_{x1} - R_{y1}$ $R_{x2} - R_{y2}$... $R_{xn} - R_{yn}$

$$\rho = \frac{1 - \frac{6 \sum d^2}{n(n^2 - 1)}}{1} \quad \text{where } n \text{ is observation} \quad -1 < \rho < 1$$

Note: the formula applied when rank are not repeated

1. Calculate the rank of co-relation coefficient of the following data

marks in ^{networks} phy: 36 43 47 28 35 50 40

mark in maths: 73 44 35 30 20 36 40

marks in network (x): 36 43 47 28 35 50 40

marks in maths (y): 73 44 35 30 20 36 40

R_x : 5 3 2 7 6 1 4

R_y : 1 2 5 6 7 4 3

$d = R_x - R_y$: 4 1 -3 1 -1 -3 1

d^2 : 16 1 9 1 1 9 1

$$\sum d = 38 \quad n = 7$$

$$\rho = \frac{1 - \frac{6(38)}{7(7^2 - 1)}}{1} = 0.3214$$

Module 4

Complex Variable

Any no in the form of $z = x + iy$ is called complex number where x is real part & y is a imaginary part

where $i = \sqrt{-1}$

Note: ① $e^{i\theta} = \cos\theta + i\sin\theta$

② $e^{-i\theta} = \cos\theta - i\sin\theta$
 ↳ exponential form ↳ imaginary

③ $\cos(i\theta) = \cosh\theta$

④ $\sin(i\theta) = i\sinh\theta$

⑤ $\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2}$

⑥ $\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$

⑦ $\sinh\theta = \frac{e^{\theta} - e^{-\theta}}{2}$

⑧ $\cosh\theta = \frac{e^{\theta} + e^{-\theta}}{2}$

Function of a complex Variable

$w = f(z) = u + iv$

A function of a complex variable defined as $w =$

$w = f(z) = u(x, y) + iu(x, y)$ is Cartesian form

$w = f(z) = u(r, \theta) + iu(r, \theta)$ polar form

where $z = x + iy$ - Cartesian

$z = re^{i\theta}$ - polar

Ex: ① $f(z) = z^2$

$$f(z) = (x + iy)^2$$

$$= x^2 + (iy)^2 + 2x(iy)$$

$$u + iv = x^2 - y^2 + 2xyi$$

$$u = x^2 - y^2, v = 2xy$$

② $f(z) = \log z$

$$= \log(re^{i\theta})$$

$$= \log r + \log e^{i\theta}$$

$$u + iv = \log r + i\theta \log e$$

$$u + iv = \log r + i\theta (1) \quad \log e = 1$$

$$u = \log r, v = \theta$$

Definition of limit

A complex valid func. $f(z)$ defined in the defining neighbourhood of the point z_0 is said to limit $z \rightarrow z_0$

$$[f(z) = z \xrightarrow{\lim} z_0] \quad f(z) \quad i.e$$

continuity

A complex valid function $f(z)$ is said to be continuous at $z = z_0$ if $f(z) = 1$

$$z \xrightarrow{\lim} z_0$$

differentiability
 A complex valued function $f(z)$ is said to be differentiable at $z = z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$ exist & a unique.

Cauchy's - Riemann eqn in Cartesian

The necessary condition that function $f(z)$ to be analytic at $z = x + iy$ is that it should be satisfies Cauchy's Riemann eqn & $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ & $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$.

Cauchy's Riemann eqn in Polar form

If $f(z) = u + iv$ is said to be analytic at the point z then there exist continuous first order partial derivatives & satisfy the eqn. $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$ & $\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$

Note: $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$ in Cartesian form

$$f'(z) = e^{-i\theta} \left[\frac{\partial u}{\partial r} + i \frac{\partial u}{\partial \theta} \right]$$

type 1: Show that z is analytic [CR eqn should be satisfy] & hence find $\frac{dw}{dz}$ or $f'(z)$

Ex: $w = \log z$, $z \neq 0$ is analytic & hence find $\frac{dw}{dz}$

$$\Rightarrow w = \log z$$

$$u + iv = \log(re^{i\theta})$$

$$u + iv = \log r + \log e^{i\theta}$$

$$u + iv = \log r + i\theta \log e$$

$$u + iv = \log r + i\theta(1)$$

$$u = \log r, \quad v = \theta$$

$$\frac{\partial u}{\partial r} = \frac{1}{r}$$

$$\frac{\partial u}{\partial \theta} = 0$$

$$\frac{\partial v}{\partial r} = 0$$

$$\frac{\partial v}{\partial \theta} = 1$$

C-Riemann $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$ & $\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$ C-Riemann satisfied & it is analytic

$$f'(z) = e^{-i\theta} \left[\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right]$$

$$= e^{-i\theta} \left[\frac{1}{r} + i(0) \right]$$

put $r=z$, $\theta=0$ using miller's thomson method.

$$f'(z) = e^{-0} \left[\frac{1}{z} + 0 \right]$$

$$\underline{f'(z) = \frac{1}{z}}$$

2. Show that $f(z) = z + e^z$ is analytic & hence find $f'(z)$

$$\hookrightarrow f(z) = z + e^z$$

$$u + iv = x + iy + e^{x+iy}$$

$$u + iv = x + iy + e^x e^{iy}$$

$$u + iv = x + iy + e^x [\cos y + i \sin y]$$

$$u + iv = (x + e^x \cos y) + i(y + e^x \sin y)$$

$$u = x + e^x \cos y, \quad v = y + e^x \sin y$$

$$\frac{\partial u}{\partial x} = 1 + e^x \cos y$$

$$\frac{\partial v}{\partial x} = 0 + e^x \sin y$$

$$\frac{\partial u}{\partial y} = 0 - e^x \sin y$$

$$\frac{\partial v}{\partial y} = 1 + e^x \cos y$$

C-R eqn

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \Rightarrow \text{C-R eqn is satisfied \& it is analytic}$$

$$f'(z) = \left[\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right]$$

$$= [(1 + e^x \cos y) + i(e^x \sin y)]$$

$$\text{put } x=z, \quad y=0$$

$$= [1 + e^z(1) + i e^z(0)]$$

$$\underline{f'(z) = 1 + e^z}$$

3. $f(z) = z^n$ is analytic & find its derivatives.

$$f(z) = z^n$$

$$u + iv = (re^{i\theta})^n$$

$$u + iv = r^n e^{in\theta}$$

$$= r^n [\cos n\theta + i \sin n\theta]$$

$$u + iv = r^n \cos n\theta + i r^n \sin n\theta$$

$$u = r^n \cos n\theta, \quad v = r^n \sin n\theta$$

$$\frac{\partial u}{\partial r} = n r^{n-1} \cos n\theta$$

$$\frac{\partial v}{\partial r} = n r^{n-1} \sin n\theta$$

$$\frac{\partial u}{\partial \theta} = r^n (-\sin n\theta) n$$

$$\frac{\partial v}{\partial \theta} = r^n \cos n\theta \cdot n$$

$$\text{C-R eqn: } \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \& \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$$

eqn is satisfied & it is analytic.

$$f'(z) = e^{-i\theta} \left[\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right]$$

$$= e^{-i\theta} [n r^{n-1} \cos n\theta + i n r^{n-1} \sin n\theta]$$

$$\text{put } r = z, \quad \theta = 0$$

$$= e^{-i0} [n z^{n-1} (1) + i (0)]$$

$$\underline{\underline{f'(z) = n z^{n-1}}}$$

4. show that $f(z) = \sin z$ is analytic & hence find $\frac{dw}{dz}$ derivatives of $f(z)$ & $f'(z)$

$$\rightarrow f(z) = \sin z$$

$$= \sin(x + iy)$$

$$= \sin x \cosh y + i \cos x \sinh y$$

$$u + iv = \sin x \cosh y + i \cos x \sinh y$$

$$u + iv = \sin x \cosh y \quad v = \cos x \sinh y$$

$$\frac{\partial u}{\partial x} = \cos x \cosh y$$

$$\frac{\partial v}{\partial x} = -\sin x \sinh y$$

$$\frac{\partial u}{\partial y} = \sin x \sinh y$$

$$\frac{\partial v}{\partial y} = \cos x \cosh y$$

C-Riemann eqn

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

eqn is satisfied & it is analytic.

$$f'(z) = \left[\frac{\partial u}{\partial x} + i \frac{\partial u}{\partial y} \right]$$
$$= [\cos x \cosh y - i \sin x \sinh y]$$

put $x=z, y=0$

$$f'(z) = \cos z(1) - i(0)$$

$$f'(z) = \underline{\underline{\cos z}}$$

type 2

Construct ^{the} analytic function $f(z)$ given real part or imaginary part.

step 1: $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$ using C-R eqn.

$$f'(z) = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \quad \text{or} \quad f'(z) = \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x}$$

real u
imag v

step 2: put $x=z, y=0$ using miller thomson method

step 3: integration w.r.t z

Determine the analytic function whose part is $y + e^x \cos y$

$$\hookrightarrow u = y + e^x \cos y$$

$$\frac{\partial u}{\partial x} = 0 + e^x \cos y, \quad \frac{\partial u}{\partial y} = 1 - e^x \sin y$$

consider $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$ using C-R eqn. $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$

$$f'(z) = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$$

$$= (e^x \cos y) - i(1 - e^x \sin y)$$

put $x=z, y=0$

$$f'(z) = e^z(1) - i(1 - e^z(0))$$

$$f'(z) = e^z - i$$

integrate

$$f(z) = \int (e^z - i) dz$$

$$\Rightarrow f(z) = \underline{\underline{e^z - iz + C}}$$

2. Find the analytic fun whose imaginary part is $e^x(x \sin y + y \cos y)$

$$\rightarrow v = e^x(x \sin y + y \cos y)$$

$$\frac{\partial v}{\partial x} = e^x(1 \sin y + 0) + (x \sin y + y \cos y)e^x$$

$$\frac{\partial v}{\partial y} = e^x(x \cos y + y(-\sin y) + \cos y(1))$$

$$= e^x(x \cos y - y \sin y + \cos y)$$

consider $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$ using C-R eqn.

$$f'(z) = \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x}$$

$$f'(z) = e^x(x \cos y - y \sin y + \cos y) + i \{ e^x(\sin y) + (x \sin y + y \cos y)e^x \}$$

$$\text{put } x = z, y = 0$$

$$f'(z) = e^z(z(1) - 0 + 1) + i(0 + 0 + 0)$$

$$f'(z) = e^z(z+1)$$

Integration

$$f(z) = \int e^z(z+1) dz$$

$$= (z+1)e^z - (1+0)e^z + C$$

$$= e^z(z+1-1) + C$$

$$f(z) = \underline{e^z z} + C$$

3. Find the analytic $f(z)$ given by $u = e^{2x}(x \cos 2y - y \sin 2y)$

$$\rightarrow u = e^{2x}(x \cos 2y - y \sin 2y)$$

$$\frac{\partial u}{\partial x} = e^{2x}(1 \cos 2y - 0) + (x \cos 2y - y \sin 2y)e^{2x} \cdot 2$$

$$\frac{\partial u}{\partial y} = e^{2x}[-x \sin 2y \cdot 2 - (y \cos 2y \cdot 2 + \sin 2y(1))]$$

$$= e^{2x}[-2x \sin 2y - 2y \cos 2y - \sin 2y]$$

consider $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$ using C-R eqn

$$= \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$$

$$f'(z) = e^{2x} (\cos 2y) + (x \cos 2y - y \sin 2y) e^{2x} \cdot 2 - i e^{2x} \{-2x \sin 2y - 2y \cos 2y - \sin 2y\}$$

put $x=0, y=0$

$$f'(z) = e^{2z} (1) + (z-0) e^{2z} \cdot 2 + i(0)$$

$$f'(z) = e^{2z} (1+2z)$$

integrate

$$f(z) = \int e^{2z} (1+2z) dz = (1+2z) \frac{e^{2z}}{2} - (0+2) \frac{e^{2z}}{4} + C$$

$$= (1+2x) \frac{e^{2z}}{2} - \frac{e^{2z}}{2} + C$$

$$= \frac{e^{2z}}{2} + \frac{2ze^{2z}}{2} - \frac{e^{2z}}{2} + C = \underline{\underline{ze^{2z} + C}}$$

4. Find the analytic $f(z) = u+iv$ given that $x^2 - y^2 = \frac{x}{x^2+y^2}$
given that

$$\hookrightarrow u = x^2 - y^2 + \frac{x}{x^2+y^2}$$

Quotient rule

$$\frac{\partial u}{\partial x} = 2x - 0 + \left[\frac{(x^2+y^2)(1) - x(2x+0)}{(x^2+y^2)^2} \right] = \frac{x du - u dx}{v^2}$$

$$\frac{\partial u}{\partial x} = 2x + \frac{y^2 - x^2}{(x^2+y^2)^2} \quad \frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$$

$$\frac{\partial u}{\partial y} = 0 - 2y + \left(\frac{-x}{(x^2+y^2)^2} \cdot (0+2y) \right) = -2y - \frac{2xy}{(x^2+y^2)^2}$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial y}$$

$$= \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \text{ when using C-R eq}$$

$$= \left(2x + \frac{y^2 - x^2}{(x^2+y^2)^2} \right) - i \left(-2y - \frac{2xy}{(x^2+y^2)^2} \right)$$

put $x=z, y=0$

$$= \left(2z + \frac{0 - z^2}{(z^2+0)^2} \right) - i(0-0)$$

$$f'(z) = 2z - \frac{1}{z^2}$$

Integ rate

$$f(z) = \int (2z - \frac{1}{z^2}) dz = \frac{2z^2}{2} - \frac{z^{-2+1}}{-2+1} \quad \text{formula}$$

$$= z^2 + \frac{z^{-1}}{1} = z^2 + \frac{1}{z} \Rightarrow \int \frac{1}{x^n} = x^{-n} = \frac{x^{-n+1}}{-n+1}$$

5. construct the analytic fun whose imaginary part is given by

given $v = \frac{-\sin \theta}{r}$ polar: $e^{-i\theta}$

$$\frac{\partial v}{\partial r} = \frac{\sin \theta}{r^2}, \quad \frac{\partial v}{\partial \theta} = \frac{-\cos \theta}{r}$$

consider $f'(z) = e^{-i\theta} \left[\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right]$

$$= e^{-i\theta} \left[\frac{1}{r} \frac{\partial v}{\partial \theta} + i \frac{\partial v}{\partial r} \right] \text{ using C-R eqn.}$$

$$= e^{-i\theta} \left[\frac{1}{r} \left(\frac{-\cos \theta}{r} \right) + i \left(\frac{\sin \theta}{r^2} \right) \right]$$

put $r=z, \theta=0$

$$f'(z) = e^{-i0} \left[\frac{1}{z} \left(\frac{-1}{z} \right) + i \right]$$

$$f'(z) = \frac{-1}{z^2}$$

integrate

$$f(z) = \int \frac{-1}{z^2} dz = \frac{-z^{-2+1}}{-2+1} + C = \frac{1}{z} + C$$

6. Find the analytic fun. $f(z) = u + iv$ given $v = \left(r - \frac{1}{r}\right) \sin \theta, r \neq 0$

$$\Rightarrow v = \left(r - \frac{1}{r}\right) \sin \theta$$

$$\frac{\partial v}{\partial r} = \left(1 + \frac{1}{r^2}\right) \sin \theta \quad \frac{\partial v}{\partial \theta} = \left(r - \frac{1}{r}\right) \cos \theta$$

consider $f'(z) = e^{-i\theta} \left[\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right]$

$$= e^{-i\theta} \left[\frac{1}{r} \frac{\partial v}{\partial \theta} + i \frac{\partial v}{\partial r} \right] \text{ using C-R eqn}$$

$$= e^{-i\theta} \left[\frac{1}{r} \left(r - \frac{1}{r}\right) \cos \theta + i \left(1 + \frac{1}{r^2}\right) \sin \theta \right]$$

put $z=2$, $\theta=0$

$$f'(z) = e^{-0} \left(\frac{1}{2} \left(2 - \frac{1}{2} \right) (1) + 0 \right)$$

$$f'(z) = 1 - \frac{1}{2z^2}$$

Integrate

$$f(z) = \int \left(1 - \frac{1}{2z^2} \right) dz \Rightarrow z + \frac{1}{2z} + C$$

Harmonic equation

If $f(z) = u + iv$ is analytic function then u & v both satisfy 2 dimensional Laplace eqn. which are all called "harmonic equation"

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

Diff wrt x

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial x \partial y} \quad \text{--- (1)}$$

Diff wrt y

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial y^2} \quad \text{--- (2)}$$

1 + 2

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

similarly $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$

Polar form

$$\textcircled{1} \quad \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$

$$\textcircled{2} \quad \frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} + \frac{1}{r^2} \frac{\partial^2 v}{\partial \theta^2} = 0$$

1. Show that $v = 2xy - 2x + 4y$ is harmonic & find its harmonic conjugate & also find $f(z)$

$$\rightarrow \frac{\partial v}{\partial x} = 2y + 2$$

$$\frac{\partial v}{\partial y} = 2x + 4$$

$$\frac{\partial^2 v}{\partial x^2} = 0$$

$$\frac{\partial^2 v}{\partial y^2} = 0$$

$$\therefore \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

It is harmonic

consider

$$\frac{\partial v}{\partial x} = 2y - 2$$

using C-R eqn

$$-\frac{\partial u}{\partial y} = 2y - 2$$

integrate using

$$u = -\int (2y - 2) dy$$

$$u = -\frac{2y^2}{2} + 2y + C$$

$$u = -y^2 + 2y + C$$

$$u = -y^2 + 2y + x^2 + 4x$$

$$f(z) = u + iv$$

$$f(z) = (-y^2 + 2y + x^2 + 4x) + i(2xy - 2x + 4y)$$

put $x = z$, $y = 0$

$$f(z) = (z^2 + 4z) + i(-2z)$$

Q. $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is harmonic & find its harmonic conjugate and also find $f(z)$

$$\therefore \frac{\partial u}{\partial x} = 3x^2 - 3y^2 + 6x$$

$$\frac{\partial^2 u}{\partial x^2} = 6x + 6$$

$$\frac{\partial u}{\partial y} = -6xy - 6y$$

$$\frac{\partial^2 u}{\partial y^2} = -6x - 6$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = (6x + 6) + (-6x - 6) \Rightarrow 0$$

$\therefore$ it is harmonic

consider

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 + 6x$$

C-R eqn

$$\frac{\partial v}{\partial y} = 3x^2 - 3y^2 + 6x$$

integrate w.r.t 'y'

$$v = \int (3x^2 - 3y^2 + 6x) dy$$

$$\frac{\partial u}{\partial y} = -6xy - 6y$$

$$-\frac{\partial v}{\partial x} = -6xy - 6y$$

$$v = \int (6xy + 6y) dx$$

$$v = 3x^2y - \frac{3y^3}{3} + 6xy$$

$$v = 3x^2y - y^3 + 6xy$$

$$v = \frac{6x^2y}{2} + 6xy$$

$$v = 3x^2y + 6xy$$

$$\therefore v = 3x^2y - y^3 + 6xy$$

$$f(z) = u + iv$$

$$f(z) = (x^3 - 3xy^2 + 3x^2 - 3y^2 + 1) + i(3x^2y - y^3 + 6xy)$$

$$\text{put } x=z, y=0$$

$$f(z) = z^3 + 3z^2 + 1$$

3. Show that $u = e^{2x}(x \cos 2y - y \sin 2y)$ is harmonic & find its analytic function.

$$\Rightarrow \frac{\partial u}{\partial x} = e^{2x}(\cos 2y - 0) + (x \cos 2y - y \sin 2y)e^{2x} \cdot 2$$

$$\frac{\partial^2 u}{\partial x^2} = 2e^{2x}(\cos 2y) + (x \cos 2y - y \sin 2y)e^{2x} \cdot 4 + e^{2x} \cdot 2(\cos 2y - 0)$$

$$\frac{\partial^2 u}{\partial x^2} = e^{2x}(4 \cos 2y + 4x \cos 2y - 4y \sin 2y)$$

$$\frac{\partial u}{\partial y} = e^{2x}(-x \cdot 2 \sin 2y - (y \cos 2y + \sin 2y))$$

$$\frac{\partial u}{\partial y} = e^{2x}[-2x \sin 2y - 2y \cos 2y - \sin 2y]$$

$$\frac{\partial^2 u}{\partial y^2} = e^{2x}[-4x \cos 2y - 2[(y(-\sin 2y \cdot 2) + \cos 2y)] - [2 \cos 2y]]$$

$$= e^{2x}[-4x \cos 2y + 4y \sin 2y - 4 \cos 2y]$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$f'(z) = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$$

$$= e^{2x}[\cos 2y + 2x \cos 2y - 2y \sin 2y] - i e^{2x}(-2x \sin 2y - 2y \cos 2y - \sin 2y)$$

$$\text{put } x=z, y=0$$

$$f'(z) = e^{2z}(1 + 2z - 0) - i(0)$$

$$f(z) = \int e^{2z}(1 + 2z) dz$$

$$(1+z^2) \frac{e^{2z}}{2} - (0+2) \frac{e^{2z}}{4} + C$$

$$= \frac{e^{2z}}{2} + ze^{2z} - \frac{e^{2z}}{2} + C = \underline{\underline{2e^z + C}}$$

H. show that $u = \frac{1}{r} \cos \theta$ is harmonic & find its harmonic conjugate. & also find $f(z)$

$$u = \frac{1}{r} \cos \theta$$

$$\frac{\partial u}{\partial \theta} = \frac{1}{r} (-\sin \theta)$$

$$\frac{\partial^2 u}{\partial \theta^2} = \frac{1}{r} (-\cos \theta)$$

$$\frac{\partial u}{\partial r} = -\frac{1}{r^2} \cos \theta$$

$$\frac{\partial^2 u}{\partial r^2} = \frac{2}{r^3} \cos \theta$$

$$\frac{\partial^2 u}{\partial r \partial \theta} = -\frac{1}{r^2} (-\sin \theta)$$

consider

$$\frac{\partial u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}$$

$$= \frac{2}{r^3} \cos \theta + \frac{1}{r} \left(-\frac{1}{r^2} \cos \theta \right) + \frac{1}{r^2} \left(-\frac{1}{r} \cos \theta \right)$$

$$= \frac{2}{r^3} \cos \theta - \frac{1}{r^3} \cos \theta - \frac{1}{r^3} \cos \theta$$

$$= \frac{2}{r^3} \cos \theta - \frac{2}{r^3} \cos \theta = 0 \quad \therefore u \text{ is harmonic}$$

consider

$$\frac{\partial u}{\partial \theta} = -\frac{1}{r} \sin \theta$$

C-R eqn.

$$-r \frac{\partial v}{\partial r} = -\frac{1}{r} \sin \theta$$

$$\frac{\partial v}{\partial r} = \frac{1}{r^2} \sin \theta$$

Integrate

$$v = \int \frac{1}{r^2} \sin \theta dr$$

$$v = \frac{r^{-2+1}}{-2+1} \sin \theta$$

$$v = -\frac{1}{r} \sin \theta \rightarrow \text{harmonic conjugate}$$

Consider $f(z) = u + iv$

$$f(z) = \frac{1}{r} \cos \theta + i \left(-\frac{1}{r} \sin \theta \right)$$

put $r = z, \theta = 0$

$$f(z) = \frac{1}{z}$$

$$\frac{\partial u}{\partial \theta} = -\frac{1}{r^2} \cos \theta$$

$$\frac{1}{r} \frac{\partial v}{\partial \theta} = -\frac{\cos \theta}{r^2}$$

$$\frac{\partial v}{\partial \theta} = -\frac{\cos \theta}{r}$$

$$v = -\int \frac{\cos \theta}{r} d\theta$$

$$v = -\frac{\sin \theta}{r}$$

5. show that $u = (r - \frac{1}{r}) \cos \theta$ is harmonic & find its harmonic conjugate

- gate & also find $f(z)$

$$\rightarrow \frac{\partial u}{\partial r} = (1 + \frac{1}{r^2}) \cos \theta$$

$$\frac{\partial^2 u}{\partial r^2} = (0 - \frac{2}{r^3}) \cos \theta$$

$$\frac{\partial u}{\partial \theta} = (r - \frac{1}{r}) (-\sin \theta)$$

$$\frac{\partial^2 u}{\partial \theta^2} = (r - \frac{1}{r}) (-\cos \theta)$$

consider

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}$$

$$= -\frac{2}{r^3} \cos \theta + \frac{1}{r} (1 + \frac{1}{r^2}) \cos \theta + \frac{1}{r^2} (r - \frac{1}{r}) (-\cos \theta)$$

$$= -\frac{2}{r^3} \cos \theta + \frac{1}{r} \cos \theta + \frac{1}{r^3} \cos \theta - \frac{1}{r} \cos \theta + \frac{1}{r^3} \cos \theta$$

$$= 0$$

$\therefore u$ is harmonic

$$\frac{\partial u}{\partial r} = (1 + \frac{1}{r^2}) \cos \theta$$

$$\left[\frac{1}{r} \frac{\partial v}{\partial \theta} = (1 + \frac{1}{r^2}) \cos \theta \right]$$

$$\frac{\partial v}{\partial \theta} = (r + \frac{r}{r^2}) \cos \theta$$

$$v = \int (r + \frac{r}{r^2}) \cos \theta \, d\theta$$

$$v = (r + \frac{1}{r}) (\sin \theta)$$

$$\frac{\partial u}{\partial \theta} = (r - \frac{1}{r}) (-\sin \theta)$$

$$-r \frac{\partial v}{\partial r} = (r - \frac{1}{r}) (-\sin \theta)$$

$$\frac{\partial u}{\partial r} = (-1 + \frac{1}{r^2}) \sin \theta$$

$$v = - \int (-1 + \frac{1}{r^2}) \sin \theta \, dr$$

$$v = -(-r + \frac{1}{r}) \sin \theta$$

$$v = (r + \frac{1}{r}) \sin \theta \text{ is harmonic}$$

$$f(z) = u + iv$$

$$f(z) = (r - \frac{1}{r}) \cos \theta + i (r + \frac{1}{r}) \sin \theta$$

$$\text{put } r = z, \theta = 0$$

$$f(z) = (z - \frac{1}{z})$$

Prove that the function $u = \sin x \cosh y + 2 \cos x \sinh y + x^2 - y^2$ is harmonic & find its analytic function.

$$u = \sin x \cosh y + 2 \cos x \sinh y + x^2 - y^2 + 4xy$$

$$\frac{\partial u}{\partial x} = \cos x \cosh y - 2 \sin x \sinh y + 2x - 0 + 4y$$

$$\frac{\partial^2 u}{\partial x^2} = -\sin x \cosh y - 2 \cos x \sinh y + 2 - 0 \quad \text{--- (1)}$$

$$\frac{\partial u}{\partial y} = \sin x \sinh y + 2 \cos x \cosh y - 2y + 4x$$

$$\frac{\partial^2 u}{\partial y^2} = \sin x \cosh y + 2 \cos x \sinh y - 2 \quad \text{--- (2)}$$

adding (1) & (2)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \therefore u \text{ is harmonic function.}$$

$$\frac{\partial}{\partial y} \cosh y = \sinh y \quad \frac{\partial}{\partial y} (\sinh y) = \cosh y$$

$$\text{consider } f(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial z} \\ = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} \quad \text{C-R eqn.}$$

$$= (\cos x \cosh y - 2 \sin x \sinh y + 2x - 4y) - i(\sin x \sinh y + 2 \cos x \cosh y)$$

put $x=z, y=0$

$$f(z) = (\cos z - 0 + 2z - 0) - i(2 \cos z + 4z)$$

integrate

$$f(z) = \int (\cos z + 2z) - i(2 \cos z + 4z) dz$$

$$= \sin z + \frac{2z^2}{2} - i(2 \sin z + \frac{4z^2}{2}) + C$$

$$= \sin z + z^2 - i(2 \sin z + 2z^2) + C$$

Ans

Given $u-v = (x-y)(x^2+4xy+y^2)$ find the analytic function.

$$u-v = x^3 + 4x^2y + xy^2 - yx^2 - 4xy^2 - y^3$$

$$u-v = x^3 + 3x^2y - 3xy^2 - y^3$$

Diff wrt 'x' partially

$$\frac{\partial u}{\partial x} - \frac{\partial v}{\partial x} = 3x^2 + 6xy - 3y^2 \quad \text{--- (1)}$$

Diff wrt 'y' partially

$$\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} = 0 + 3x^2 - 6xy - 3y^2 \quad \text{--- (2)}$$

using C-R eqn in (2)

$$-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 3x^2 - 6xy - 3y^2 \quad \text{--- (3)}$$

(1) + (3)

$$-2 \frac{\partial v}{\partial x} = 6x^2 - 6y^2$$

$$\frac{\partial v}{\partial x} = -3x^2 + 3y^2$$

(1) - (2)

$$2 \frac{\partial u}{\partial x} = 12xy$$

$$\frac{\partial u}{\partial x} = 6xy$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$[f'(z) = i(-3x^2 + 3y^2) + i(6xy)]$$

put $x = z, y = 0$

$$f'(z) = (-3z^2 + 0) + i(0)$$

$$f(z) =$$

$$f'(z) = 6xy + i(-3x^2 + 3y^2)$$

put $x = z, y = 0$

$$f'(z) = 0 + i(-3z^2 + 0)$$

$$f(z) = i \int -3z^2 dz = -\frac{3z^3}{3} + C$$

$$f(z) = -iz^3 + C$$

8. $u - v = e^x (\cos y - \sin y)$ find the analytic fun.

$$\Rightarrow \frac{\partial u}{\partial x} - \frac{\partial v}{\partial x} = e^x (\cos y - \sin y) \quad \text{--- (1)}$$

$$\frac{\partial u}{\partial y} - \frac{\partial v}{\partial y} = e^x (-\sin y - \cos y) \quad \text{--- (2)}$$

$$\text{consider } f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$f'(z) = e^x \cos y + i e^x \sin y$$

put $x = z, y = 0$

$$[f(z) = \int e^z (1) + i(0)]$$

$$f(z) = e^z + C$$

using C-R eqn.

$$-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} = e^x (-\sin y - \cos y) \quad \text{--- (1) - (2)}$$

(1) + (3)

$$-2 \frac{\partial v}{\partial x} = e^x (-2 \sin y)$$

$$\frac{\partial v}{\partial x} = e^x \sin y$$

$$2 \frac{\partial v}{\partial x} = e^x (2 \cos y)$$

$$\frac{\partial u}{\partial x} = e^x \cos y$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$= e^x \cos y + i e^x \sin y$$

$$x = z, y = 0$$

type 3

flow problems

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$|f'(z)| = \sqrt{\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2}$$

$$|f'(z)|^2 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2$$

similarly

$$|f'(z)|^2 = \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2$$

1.4 $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^2 = 4 |f'(z)|^2$ is analytic function.

Let $f(z) = u + i v$
 $|f(z)| = \sqrt{u^2 + v^2}$

Sq. on b.s

$$|f(z)|^2 = u^2 + v^2$$

diff wrt x

$$\frac{\partial}{\partial x} |f(z)|^2 = 2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x}$$

again diff

$$\frac{\partial^2}{\partial x^2} |f(z)|^2 = 2 \left[u \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + v \frac{\partial^2 v}{\partial x^2} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial x} \right]$$

$$\frac{\partial^2}{\partial x^2} |f(z)|^2 = 2 \left[u \frac{\partial^2 u}{\partial x^2} + \left(\frac{\partial u}{\partial x}\right)^2 + v \frac{\partial^2 v}{\partial x^2} + \left(\frac{\partial v}{\partial x}\right)^2 \right] \rightarrow \text{①}$$

similarly

$$\frac{\partial^2}{\partial y^2} |f(z)|^2 = 2 \left[u \frac{\partial^2 u}{\partial y^2} + \left(\frac{\partial u}{\partial y}\right)^2 + v \frac{\partial^2 v}{\partial y^2} + \left(\frac{\partial v}{\partial y}\right)^2 \right] \rightarrow \text{②}$$

using C-R eqn

$$\frac{\partial^2}{\partial y^2} |f(z)|^2 = 2 \left[u \frac{\partial^2 u}{\partial y^2} + \left(-\frac{\partial u}{\partial x}\right)^2 + v \frac{\partial^2 v}{\partial y^2} + \left(\frac{\partial v}{\partial x}\right)^2 \right] \rightarrow \text{③}$$

① + ②

$$\frac{\partial^2}{\partial x^2} |f(z)|^2 + \frac{\partial^2}{\partial y^2} |f(z)|^2 = 2 \left[u \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + 2 \left(\frac{\partial u}{\partial x} \right)^2 + v \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + 2 \left(\frac{\partial v}{\partial x} \right)^2 \right]$$

$$f'(z) = e^z + i(0)$$

$$f(z) = \int e^z dz$$

$$f(z) = \underline{e^z + C}$$

u, v are harmonic
so zero

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^2 = 2 \left[u(0) + 2\left(\frac{\partial u}{\partial x}\right)^2 + v(0) + 2\left(\frac{\partial v}{\partial x}\right)^2 \right]$$

$$= 4 \left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 \right]$$

$$= 4 |f'(z)|^2$$

Example

2. $f(z) = u + iv$ is analytic fun then proof that

$$\left(\frac{\partial}{\partial x} |f(z)|\right)^2 + \left(\frac{\partial}{\partial y} |f(z)|\right)^2 = |f'(z)|^2$$

$$\rightarrow f(z) = u + iv$$

$$|f(z)| = \sqrt{u^2 + v^2}$$

$$\frac{\partial}{\partial x} |f(z)| = \frac{1}{\sqrt{u^2 + v^2}} \left[u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x} \right]$$

Sq on b.s

$$\left(\frac{\partial}{\partial x} |f(z)|\right)^2 = \frac{1}{u^2 + v^2} \left[u^2 \left(\frac{\partial u}{\partial x}\right)^2 + v^2 \left(\frac{\partial v}{\partial x}\right)^2 + 2uv \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} \right] \quad \text{--- (1)}$$

Similar

$$\left(\frac{\partial}{\partial y} |f(z)|\right)^2 = \frac{1}{u^2 + v^2} \left[u^2 \left(\frac{\partial u}{\partial y}\right)^2 + v^2 \left(\frac{\partial v}{\partial y}\right)^2 + 2uv \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} \right] \quad \text{--- (2)}$$

using C-R eqn (2)

$$\left(\frac{\partial}{\partial y} |f(z)|\right)^2 = \frac{1}{u^2 + v^2} \left[u^2 \left(-\frac{\partial v}{\partial x}\right)^2 + v^2 \left(\frac{\partial u}{\partial x}\right)^2 + 2uv \left(-\frac{\partial v}{\partial x}\right) \left(\frac{\partial u}{\partial x}\right) \right]$$

(1) + (2)

$$\left(\frac{\partial}{\partial x} |f(z)|\right)^2 + \left(\frac{\partial}{\partial y} |f(z)|\right)^2 = \frac{1}{u^2 + v^2} \left[(u^2 + v^2) \left(\frac{\partial u}{\partial x}\right)^2 + (u^2 + v^2) \left(\frac{\partial v}{\partial x}\right)^2 \right]$$

$$= \frac{1}{u^2 + v^2} (u^2 + v^2) \left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 \right]$$

$$= |f'(z)|^2$$

Application of the flow problems

$$f(z) = u + iv, \quad f(z) = \phi + i\psi$$

where ϕ = potential function / velocity potential

ψ = Stream fun

$f(z)$ = complex potential

Ex 1 In electro magnetic field in xy plane is given by the potential function $\phi = x^2 - y^2$, find the stream function & also find complex potential.

→ $\phi = x^2 - y^2 = u$

$$\frac{\partial \phi}{\partial x} = \frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial \phi}{\partial y} = \frac{\partial u}{\partial y} = -2y$$

apply C-R eqn.

$$\frac{\partial v}{\partial y} = 2x$$

integrate

$$v = \int 2x \, dy$$

$$v = 2xy$$

$$f(z) = u + i v$$

$$f(z) = (x^2 - y^2) + i(2xy)$$

put $x = z$; $y = 0$

$$f(z) = \underline{z^2}$$

note: In a yield flow problem $\phi =$ isothermal $\psi =$ yield flow lines

* In hydrodynamics $\phi =$ velocity potential $\psi =$ stream function

* In electrostatics $\phi =$ strain function $\psi =$ stress lines

2) In a 2 dimensional fluid flow the stream fun.

$\psi = \tan^{-1}(y/x)$ find ϕ , $f(z)$

→ $\psi = v = \tan^{-1}(y/x)$

$$\frac{\partial v}{\partial x} = \frac{1}{1 + (y/x)^2} \cdot (-y/x^2) = \frac{-y}{x^2 + y^2} \quad \text{--- (1)}$$

$$\frac{\partial v}{\partial y} = \frac{1}{1 + (y/x)^2} \cdot \left(\frac{1}{x}\right) = \frac{x}{x^2 + y^2} \quad \text{--- (2)}$$

Apply C-R eqn in (2)

$$+ \frac{\partial u}{\partial x} = \frac{x}{x^2 + y^2}$$

$$\frac{\partial u}{\partial x} = \frac{x}{x^2 + y^2}$$

int x

$$u = \int \frac{x}{x^2 + y^2} \, dx$$

At $y = 0$ by (2)

$$= + \frac{1}{2} \int \frac{2x}{x^2 + y^2} \, dx$$

$$u = + \frac{1}{2} \log(x^2 + y^2)$$

$$f(z) = u + iv$$

$$f(z) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1}(y/x)$$

$$\text{put } x = z, y = 0$$

$$f(z) = \frac{1}{2} \log z^2 = \frac{1}{2} \log z = \log z$$

3) An electromagnetic field in xy plane is given by the potential plane $\phi = 3x^2 - y^3$, find stream fun also find complex potential.

$$\phi = 3x^2y - y^3 = u$$

$$\frac{\partial u}{\partial y} = 6xy \quad \text{--- ①}$$

$$\frac{\partial u}{\partial y} = 3x^2 - 3y^2 \quad \text{--- ②}$$

from ① C-R eqn.

$$\frac{\partial v}{\partial y} = 6xy$$

$$v = \int 6xy \, dy$$

$$v = 6x \frac{y^2}{2}$$

$$v = 3xy^2$$

from ② C-R eqn.

$$-\frac{\partial v}{\partial x} = 3x^2 - 3y^2$$

$$v = -\int (3x^2 - 3y^2) \, dx$$

$$v = -\left[\frac{3x^3}{3} - 3y^2x\right]$$

$$v = -x^3 + 3y^2x$$

$$v = -x^3 + 3xy^2$$

$$f(z) = u + iv$$

$$f(z) = (3x^2y - y^3) + (-x^3 + 3xy^2)i$$

$$\text{put } x = z, y = 0$$

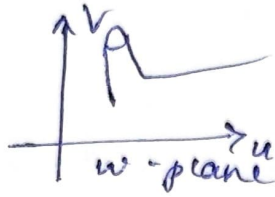
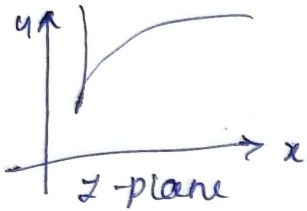
$$f(z) = 0 - z^3i$$

$$f(z) = -iz^3$$

conformal transformation

line integrals/contours

the fun $z = x + iy$ & $w = u + iv$ are the 2 plane known as z plane & w plane respectively.



Discuss the transformation

$$w = e^z$$

$$u + iv = e^{x + iy}$$

$$u + iv = e^x e^{iy}$$

$$u + iv = e^x [\cos y + i \sin y]$$

$$u = e^x \cos y \quad v = e^x \sin y$$

squaring & adding

$$u^2 + v^2 = e^{2x} (\cos^2 y + \sin^2 y) = e^{2x} = \frac{v}{\sin y}$$

$$u^2 + v^2 = e^{2x} \quad \text{--- ①}$$

$$\frac{v}{u} = \frac{e^x \sin y}{e^x \cos y}$$

$$\boxed{\frac{v}{u} = \tan y} \quad \text{--- ②}$$

case 1: put $x = c_1$ in eqn ①

$$u^2 + v^2 = e^{2c_1}$$

$$u^2 + v^2 = r^2 \quad \text{where } r = e^{c_1}$$

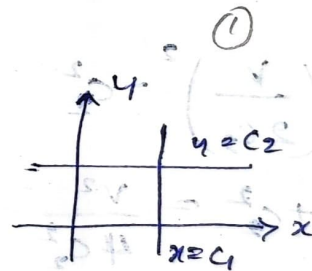
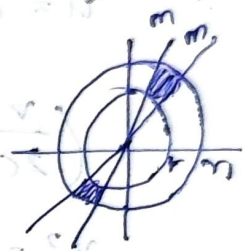
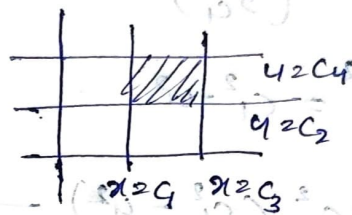
case 2: put $y = c_2$ in eqn ②

$$\frac{v}{u} = \tan c_2$$

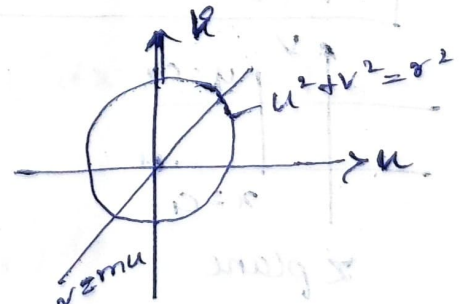
$$v = (\tan c_2) u$$

$$v = m u \quad \text{where } m = \tan c_2$$

$$w = e^z$$



z -plane



2. Discuss the transformation $w = z^2$

→ let $w = z^2$

$$u + iv = (x + iy)^2$$

$$u + iv = x^2 - y^2 + 2xyi$$

$$u = x^2 - y^2, \quad v = 2xy$$

Case 1: put $x = c_1$

$$u = c_1^2 - y^2 \quad v = 2c_1 y$$

$$\frac{v}{2c_1} = y$$

$$u = c_1^2 - \left(\frac{v}{2c_1}\right)^2$$

$$\left(\frac{v}{2c_1}\right)^2 = c_1^2 - u$$

$$v^2 = 4c_1^2 (c_1^2 - u) \quad \text{--- ①}$$

Case put $y = c_2$

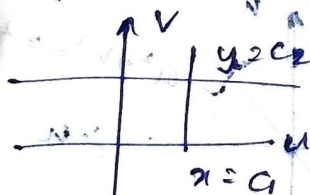
$$u = x^2 - c_2^2 \quad v = 2xc_2$$

$$\boxed{\frac{v}{2c_2} = x}$$

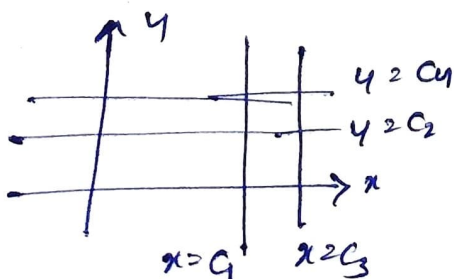
$$u = \left(\frac{v}{2c_2}\right)^2 - c_2^2$$

$$u + c_2^2 = \frac{v^2}{4c_2^2}$$

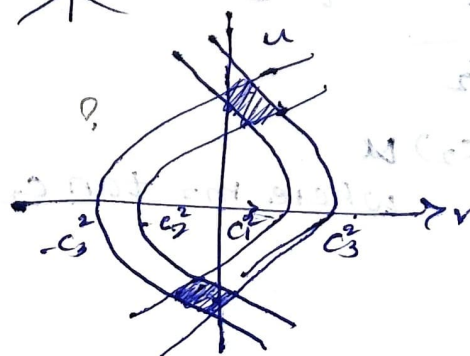
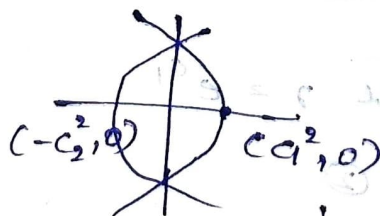
$$\boxed{4c_2^2 (u + c_2^2) = v^2}$$



z -plane



z -plane



w -plane

write only 1 pair of diagram

complex integral

line Integrals

if $f(z) = u + iv$ is a complex valued func. defined over a region R in the complex plane and C is a curve in the region then

$$\int_C f(z) dz = \int_C (u + iv)(dx + i dy)$$

$$z = x + iy$$

$$dz = dx + i dy$$

$$f(z) = u + iv$$

1. Evaluate $\int_C z^2 dz$

(1) $z = 0$ to $z = 3 + i$

(2) along the curve made up of 2 line segments

one from $z = 0$ to $z = 3 + i$ & another from $z = 3$ to $z = 3 + i$

$$\int_C z^2 dz$$

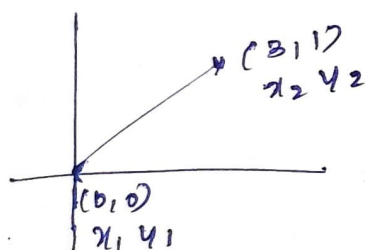
$$\int (x + iy)^2 (dx + i dy)$$

$$\int (x^2 - y^2 + 2xyi)(dx + i dy)$$

(1) along $z = 0$ to $z = 3 + i$

$$x + iy = 0 + i0$$

$$x + iy = 3 + i$$



$$\frac{(y_2 - y_1)}{(x_2 - x_1)} = \frac{y - y_1}{x - x_1}$$

$$\frac{1 - 0}{3 - 0} = \frac{y - 0}{x - 0}$$

$$\frac{1}{3} = \frac{y}{x}$$

$$y = \frac{x}{3}$$

$$dy = \frac{dx}{3}$$

from (1) $\int_{x=0}^3 (x^2 - (\frac{x}{3})^2 + 2x(\frac{x}{3})i)(dx + i \frac{dx}{3})$

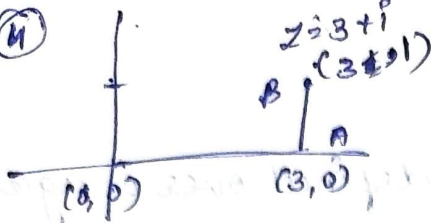
$$\int_{x=0}^3 (x^2 - \frac{x^2}{9} + \frac{2x^2}{3}i)(1 + \frac{i}{3}) dx$$

$$(1 + \frac{i}{3}) (\frac{x^3}{3} - \frac{x^3}{27} + \frac{2}{3} \frac{x^3}{3} i) \Big|_0^3$$

$$(1 + \frac{i}{3}) (\frac{27}{3} - \frac{27}{27} + \frac{2}{9} (27)i) \Rightarrow 6 + \frac{26i}{3}$$

$$\Rightarrow 6 + 8.66i$$

(4)



along OA

$$x=0, \quad y=3 \text{ \& } y=0 \quad dy=0$$

$$\begin{aligned} \text{from ①} \int_0^3 (x^2 + 2x(0)i) (dx + i(0)) \\ = \left(\frac{x^3}{3} \right)_0^3 \\ = \frac{3^3}{3} - 0 = \underline{\underline{9}} \end{aligned}$$

$$\text{along AB: } x=3 \text{ \& } y=0 \text{ to } y=1 \quad dx=0$$

$$\text{from ①} \int_0^1 (3^2 - y^2 + 2(3)y i) (0 + i dy)$$

$$= i \left[9y - \frac{y^3}{3} + 6 \frac{y^2}{2} i \right]_0^1$$

$$= i \left[9 - \frac{1}{3} + 6 \cdot \frac{1}{2} i \right]$$

$$= -3 + \frac{26}{3} i$$

$$\begin{aligned} \int_C dz &= \int_{OA} + \int_{AB} = 9 + \left(-3 + \frac{26}{3} i \right) \\ &= 6 + \frac{26}{3} i \end{aligned}$$

$$\Rightarrow 6 + \frac{26}{3} i$$

2. Evaluate $\int_0^{1+i} (x^2 + iy) dz$ along the path ① $y=x$ ② $y=x^2$

$$\Rightarrow \int_0^{1+i} (x^2 + iy) (dx + i dy)$$

$$\text{① } y=x \\ dy=dx$$

$$= \int_0^{1+i} (x^2 + ix) (dx + i dx)$$

$$= (1+i) \int_0^1 (x^2 + ix) dx$$

$$= (1+i) \left(\frac{x^3}{3} + i \frac{x^2}{2} \right)_0^1$$

$$= (1+i) \left(\frac{1}{3} + \frac{i}{2} \right)$$

$$= \frac{1}{6} + \frac{5}{6} i$$

② along $y=x^2$
 $dy=2x dx$

$$= \int_0^1 (x^2 + ix^2)(dx + i^2 2x dx)$$

$$= \int_0^1 x^2(1+i) dx(1+2xi)$$

$$= (1+i) \int_0^1 (x^2 + 2x^3 i) dx$$

$$= (1+i) \left(\frac{x^3}{3} + \frac{2x^4}{4} i \right)_0^1$$

$$= (1+i) \left(\frac{1}{3} + \frac{i}{2} \right)$$

$$= \frac{-1}{6} + \frac{5}{6} i$$

3. evaluate integration $\int_{1-i}^{2+i} (2x + iy + 1) dz$ along (i) $x = t+1, y = 2t^2-1$
 (ii) straight line joining $1-i$ & $2+i$

$$\hookrightarrow \int_{1-i}^{2+i} (2x + iy + 1) (dx + i dy)$$

along $x = t+1, y = 2t^2-1$
 $dx = dt + 0, dy = 4t dt - 0$
 $dx = dt, dy = 4t dt$

$x=1, y=-1$ $x=2, y=1$
 $0=t$ $1=t$

$$= \int_{t=0}^1 [2(t+1) + i(2t^2-1) + 1] [dt + 4ti dt]$$

$$= \int_{t=0}^1 [2t+2 + i(2t^2-1) + 1] dt [1+4ti]$$

$$= \int_{t=0}^1 [2t+3 + i(2t^2-1)] + 8t^2 i + 12ti - (8t^3-4t) dt$$

$$= \int_{t=0}^1 [2t+3 + i(2t^2-1)] + 8t^2 i + 12ti - (8t^3-4t) dt$$

$$= \left[\frac{2t^2}{2} + 3t + i\left(\frac{2t^3}{3} - t\right) + \frac{8t^3}{3} i + 12\frac{t^2}{2} i - \left[\frac{8t^4}{4} - \frac{4t^2}{2}\right] \right]_0^1$$

$$= 1 + 3 + i\left(\frac{2}{3} - 1\right) + \frac{8}{3} i + 6i - (2 - 2)$$

$$= 4 + \frac{2i}{3} - i + \frac{8i}{3} + 6i$$

$$= 4 + \frac{25}{3} i$$

(ii) along $1-i$ to $2+i$

$$\begin{matrix} (2, 1) \\ x_2, y_2 \\ \swarrow \\ (1, -1) \\ x_1, y_1 \end{matrix}$$

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\frac{y+1}{1+1} = \frac{x-1}{2-1}$$

$$\frac{y+1}{2} = \frac{x-1}{1}$$

$$y+1 = 2x-2$$

$$y = 2x-3$$

$$dy = 2dx \rightarrow 0$$

$$\int (2x + iy + 1)(dx + i dy)$$

$$\int_1^2 [2x + i(2x-3) + 1][dx + i 2dx]$$

$$= \int_1^2 [(2x + i(2x-3) + 1)][1 + 2i] dx$$

$$= (1+2i) \left[2 \frac{x^2}{2} + i \left[2 \frac{x^2}{2} - 3x \right] + x \right]_1^2$$

$$= (1+2i) \{ 4 + i(4-6) + 2 \} - [1 + i(1-3) + 1]$$

$$= (1+2i) [(6 - 2i) - (2 - 2i)]$$

$$= (1+2i)(4) = 4 + 8i$$

4. Evaluate $\int_0^{2+i} z^2 dz$ along (i) line $y = \frac{x}{2}$ or $x = 2y$

(ii) the real axis to '2' and then to $2+i$

$$\rightarrow \int_0^{2+i} (x+iy)^2 (dx + i dy)$$

$$= \int_0^{2+i} (x^2 - y^2 + 2xyi)(dx + i dy) \quad \text{--- (2)}$$

$$\textcircled{i} y = \frac{x}{2}$$

$$2y = x$$

$$2dy = dx$$

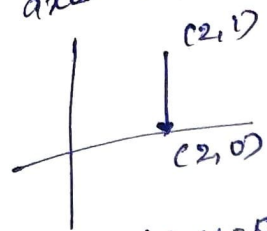
$$\int_0^1 [(2y)^2 - y^2 + 2(2y)y i][2dy + i dy]$$

$$\int_0^1 (4y^2 - y^2 + 4y^2 i) dy (2+i)$$

$$(2+i) \left[\frac{4y^3}{3} - \frac{y^3}{3} + \frac{4y^3}{3} i \right]_0^1$$

$$z(2+i) \left[\frac{4}{3} - \frac{1}{3} + \frac{4}{3}i \right] \Rightarrow \frac{2}{3} + \frac{11}{3}i$$

axis '2' to $2+i$



limit $y=0$ to $y=1$

$z=2$
 $dx=0$

$$\begin{aligned} & \int_0^1 [(2)^2 - y^2 + 2(2)yi] [0 + i dy] \\ &= i \int_0^1 (4 - y^2 + 4yi) dy \\ &= i \left[4y - \frac{y^3}{3} + \frac{4y^2}{2}i \right]_0^1 \\ &= i \left[4 - \frac{1}{3} + 2i \right] = -2 + \frac{11}{3}i \end{aligned}$$

By linear transformation

The bilinear transformation is used in digital signal processing the discrete time control theory to transform continuous time system also used in computer programs & its applications.

def: The transformation of the form $w = \frac{az+b}{cz+d}$ & $ad-bc \neq 0$ is called bilinear transform. discovered by Mobius in 1712 & it is also called Mobius transformation.

alternate form

let w is a image of z under the bilinear transformation we have $(w, w_1, w_2, w_3) = (z, z_1, z_2, z_3)$

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

1. Find the bilinear transformation which maps the point $z=1, i, -1$ into $w=i, 0, -i$

$$\hookrightarrow w_1=i \quad w_2=0 \quad w_3=-i \quad z_1=1 \quad z_2=i \quad z_3=-1$$

WKT

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w-i)(0+i)}{(i-0)(-i-w)} = \frac{(z-1)(i+1)}{(1-i)(-1-z)}$$

$$\frac{(w-i)}{-i-w} = \frac{z-1}{(-1-z)} \frac{(i+1)}{(1-i)}$$

$$\left(\frac{w-i}{-i-w} \right) = \left(\frac{z-1}{-1-z} \right) \times \left(\frac{i+1}{1-i} \times \frac{1+i}{1+i} \right)$$

$$\frac{w-i}{-i-w} = \frac{z-1}{-1-z} \left(\frac{(1+i)^2}{1^2-i^2} \right)$$

$$\frac{w-i}{-i-w} = \left(\frac{z-1}{-1-z} \right) \left(\frac{i-i+2i}{1+1} \right)$$

$$\left(\frac{w-i}{-i-w} \right) = \left(\frac{z-1}{-1-z} \right) \left(\frac{2i}{2} \right)$$

$$(w-i)(-1-z) = (-i-w)(z-i)i$$

$$-w - zw + i + iz = (-i-w)(zi-i)$$

$$-w - zw + i + iz = (-zi^2 + i^2 - wzi + wi)$$

$$-w - zw + i + iz = z - 1 - wzi + wi$$

$$(w-i)(-1-z)(1-i) = (z-1)(i+1)(-i-w)$$

$$(-w - zw + i + iz)(1-i) = (zi + z - i - 1)(-i-w)$$

$$-w - zw + i + iz + wi + zw + 1 + z = z - zi - 1 - i - zi - w$$

$$-w + iz + zw + 1 = -zi - 1 - zi - w$$

$$-2w + 2zwi = -2zi - 2$$

$$-zw(1-zi) = -z(zi+1)$$

$$w = \frac{zi+1}{1-iz}$$

Note : If any term gives infinity in w plane or in z plane that can be written as that can be written in the form of $\frac{1}{0}$

Find bilateral trans of the mapping $z=1, i-1$
 $w=0, 1, \infty$

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$w = \frac{z-i-i}{-1-z}$$

$$\frac{(w-w_1)w_3(w_2-w_3)}{(w_1-w_2)w_3(1-\frac{w}{w_3})} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$w = \frac{(z-1)i}{-1-z}$$

$$\frac{(w-0)(\frac{1}{\infty}-1)}{(0+1)(1-\frac{w}{\infty})} = \frac{(z-1)(i+1)}{(1-i)(-1-z)}$$

$$\frac{(w)(-1)}{(-1)(1)} = \frac{z-1}{-1-z} \times \frac{i+1}{1-i}$$

$$w = \frac{z-1}{-1-z} \times \frac{i+1}{1-i} \times \frac{1+i}{1+i}$$

$$w = \frac{z-1}{-1-z} \times \frac{(1+i)^2}{1^2-i^2}$$

$$w = \frac{z-1}{-1-z} \times \frac{(1-i)(1+i)}{1+1}$$

$$w = \frac{(z-1)}{(-1-z)(-i)}$$

$$= \frac{z-1}{i+iz}$$

$$w = \frac{z-1}{i+iz}$$

3. Find bilinear trans of the mapping, $z=1, i, -1$ into $w=2, i, -2$

$$\frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w-2)(i+2)}{(2-i)(-2-w)} = \frac{(z-1)(i+1)}{(1-i)(-1-z)}$$

$$\frac{(w-2)}{(-2-w)} = \frac{(z-1)}{(-1-z)} \times \frac{i+1}{1-i} \times \frac{(2-i)}{(i+2)}$$

$$\frac{w-2}{-2-w} = \left(\frac{z-i}{i-i-2} \right) \times \left(\frac{2i+1+2-i}{i+2+1-2i} \right)$$

$$\frac{w-2}{-2-w} = \frac{z-1}{-1-z} \times \left(\frac{i+3}{3-i} \right) \Rightarrow \frac{w-2}{-2-w} = \frac{z-1}{-1-z} \times \frac{i+3}{3-i}$$

$$(w-2)(-1-z)(3-i) = (z-1)(i+3)(-2-w)$$

$$\begin{aligned}
 (-w - zw + 2 + 2z)(3-i) &= (2i + 3z - i - 3)(-2-w) \\
 -3w - 3zw + 6 + 6z + wi + 2wi &= -2i - 2z - 2zi - 6z + 3i \\
 + 6 - 2iw - 3zw + 6w + 3w & \\
 -6w + 22wi &= -12z + 4i \\
 w(-6 + 22i) &= -12z + 4i
 \end{aligned}$$

$$w = \frac{-12z - 4i}{22i - 6} = -\frac{6z + 2i}{2i - 3} = \frac{6z + 2i}{3 - 2i}$$

4. Find bilinear tras 0 - i - 1 Z plane on to the point i, 1, 0 w plane

$$\rightarrow \frac{(w - w_1)(w_2 - w_3)}{(w_1 - w_2)(w_3 - w)} = \frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)}$$

$$\frac{(w - i)(1 - 0)}{(i - 1)(0 - w)} = \frac{(z - 0)(-i + 1)}{(-i + 1)(-1 - z)}$$

$$\frac{(w - i)}{(i - 1)(-w)} = \frac{z(-i + 1)}{i(-1 - z)}$$

$$\frac{(w - i)}{(-w)} = \frac{z(-i + 1)(i - 1)}{i(-1 - z)}$$

$$\frac{w - i}{-w} = \frac{z(1 + i + i - 1)}{i(-1 - z)}$$

$$\frac{(w - i)}{-w} = \frac{z(2i)}{i(-1 - z)}$$

$$\frac{(w - i)}{-w} = \frac{z(2i)}{i(-1 - z)}$$

$$(w - i)(-1 - z) = z(2(-w))$$

$$-w - zw + i + zi = -2zw$$

$$-w + zw + i + zi = 0$$

$$-w + zw = -i - zi$$

$$w(-1 + z) = -i - zi$$

$$w = \frac{-i - zi}{z - 1}$$

$$w = \frac{zi + i}{1 - z}$$

if u able write or stop there

$$w = \frac{zi + i}{1 - z}$$

$$w = \frac{z + 1}{2i - 1}$$

5. Find the bilinear transform which maps the point $z = 0, 1, \infty$ into the point $w = -5, -1, 3$ respectively.

$$\Rightarrow \frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w+5)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1) \cancel{z_3} \left(\frac{z_2}{z_3} - 1 \right)}{(z_1-z_2) \cancel{z_3} \left(1 - \frac{z}{z_3} \right)}$$

$$\frac{(w+5)(-1-3)}{(-5+1)(3-w)} = \frac{(z-0) \left(\frac{1}{\infty} - 1 \right)}{(0-1) \left(1 - \frac{z}{\infty} \right)}$$

$$\frac{(w+5)(-4)}{(-4)(3-w)} = \frac{z(0-1)}{(-1)(1-0)}$$

$$\frac{w+5}{3-w} = \frac{z}{1}$$

$$w+5 = 3z-zw$$

$$w+zw = 3z-5$$

$$w(1+z) = 3z-5$$

$$w = \frac{3z-5}{1+z}$$

5. Find the bilinear transform $z = -1, i, 1$ onto $w = 1, i, -1$

$$\Rightarrow \frac{(w-w_1)(w_2-w_3)}{(w_1-w_2)(w_3-w)} = \frac{(z-z_1)(z_2-z_3)}{(z_1-z_2)(z_3-z)}$$

$$\frac{(w-1)(i+1)}{(1-i)(-1-w)} = \frac{(z+1)(i-1)}{(1+i)(1-z)}$$

$$\frac{(w-1)}{-1-w} = \frac{(z+1)}{1-z} \times \frac{i-1}{-1-i} \times \frac{1-i}{1+i}$$

$$\frac{w-1}{-1-w} = \frac{z+1}{1-z} \times \frac{1+i-i-i^2}{-1-i+i-i^2}$$

$$\left[\frac{w-1}{-1-w} = \frac{z+1}{1-z} \left(\frac{1+i}{-2i} \right) \right]$$

$$\left[\frac{(w-1)}{(-1-w)} = \frac{z+1}{1-z} \left[\frac{(1+i)i}{2} \right] \right]$$

$$\frac{w-1}{-1-w} = \frac{z+1}{1-z} (-1)$$

$$\frac{w-1}{-1-w} = \frac{-z-1}{1-z}$$

$$(w-1)(1-z) = (-z-1)(-1-w)$$

$$\cancel{w} - zw - 1 + z = \cancel{z} + zw + 1 + \cancel{w}$$

$$-zw - 1 = zw + 1$$

$$-2zw = 2$$

$$w = \underline{\underline{-\frac{1}{z}}}$$

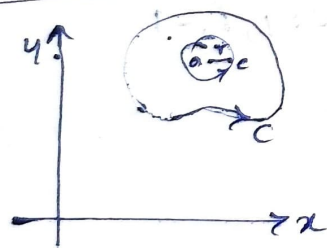
Cauchy's theorem

If $f(z)$ is analytic at all the points inside & on a simple closed curve C then integration $\oint_C f(z) dz = 0$

Cauchy's integral formula

If $f(z)$ is analytic inside & closed curve C and if a is a point within C then $\text{integ } f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a} dz$

$$\boxed{\int_C \frac{f(z)}{z-a} dz = 2\pi i f(a)}$$



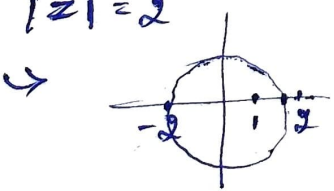
Generalize Cauchy's integral formula

If $f(z)$ is analytic inside on a ^{simple} closed curve C & a is a point within C then $\int_C \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(a)$

7m - 2 questions (cont) line integral

1. Evaluate integration $\int_C \frac{z}{(z-1)(z+2)} dz$ where C is a circle

$$|z| = 2$$



$z = 1, -2$ are lie inside

$$\text{then } \frac{z}{(z-1)(z+2)} = \frac{A}{z-1} + \frac{B}{z+2}$$

$$1 = A(z+2) + B(z-1)$$

put $z=1$, $1 = A(1+2) + 0$ $A = 1/3$

put $z=-2$

$1 = A(-2+2) + B(-2-1)$

$-\frac{1}{3} = B$

$\frac{1}{(z-1)(z+2)} = \frac{1/3}{z-1} + \frac{-1/3}{z+2}$

$\int_C \frac{z}{(z-1)(z+2)} dz = \frac{1}{3} \int_C \frac{1}{z-1} dz - \frac{1}{3} \int_C \frac{1}{z+2} dz$

$= \frac{1}{3} 2\pi i f(1) - \frac{1}{3} 2\pi i f(-2)$

$= \frac{1}{3} 2\pi i (1) - \frac{1}{3} 2\pi i (-2) = 6\pi i / 3 = 2\pi i$

$f(z) = z$

$f(1) = 1$

$f(-2) = -2$

2. Evaluate $\int_C \frac{z^2 - z + 1}{z-1} dz$ where C is a circle $|z|=1$

(i) $|z|=1/2$

$\hookrightarrow$ (i) $z=1$ on the circle $|z|=1$



$\int_C \frac{z^2 - z + 1}{z-1} dz = 2\pi i f(1)$
 $= 2\pi i (1)$

$f(z) = z^2 - z + 1$
 $f(1) = 1 - 1 + 1 = 1$

(ii) $z=1$ are out of $|z|=1/2$

$\therefore \int_C \frac{z^2 - z + 1}{z-1} dz = 0$



outside the circle
 so it is zero

3. Evaluate $\int_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$ where $C: |z|=3$

$\hookrightarrow z=1, 2$ lies inside of $|z|=3$



consider

$\frac{1}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$

$1 = A(z-2) + B(z-1)$

put $z=2$

$1 = A(0) + B(2-1)$

$\boxed{1=B}$

put $z=1$

$1 = A(1-2)$

$\boxed{A=-1}$

$$\int \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz = -1 \int \frac{1}{z-1} dz + 1 \int \frac{1}{z-2} dz$$

$$= (-1) 2\pi i f(1) + (1) 2\pi i f(2)$$

$$= -2\pi i (-1) + 2\pi i (1) = 4\pi i$$

$$f(z) = \sin \pi z^2 + \cos \pi z^2$$

$$f(1) = \sin \pi (1) + \cos \pi (1)$$

$$= 0 - 1$$

$$= -1$$

$$f(2) = \sin \pi (2)^2 + \cos \pi (2)^2$$

$$= \sin 4\pi + \cos 4\pi$$

$$= 0 + 1 = 1$$

4. Evaluate $\int_C \frac{z}{z^2 - 3z + 2} dz$ where $|z-2| = 1/2$

$$|(1+i4) - 2| = 1/2$$

$$|(1-2) + i(4-0)| = 4/2$$

$$\text{Center } (2,0) \text{ radius } = 1/2$$

$$\rightarrow z^2 - 3z + 2 = 0$$

$$z^2 - 2z - z + 2 = 0$$

$$z(z-2) - 1(z-2) = 0$$

$$(z-1)(z-2) = 0$$

$$\int_C \frac{z}{(z-1)(z-2)} dz$$

Since $z=1$ outside & $z=2$ inside

$$\int \frac{z}{z-2} dz = 2\pi i f(2)$$

$$= 2\pi i (2)$$

$$= 4\pi i$$

$$f(z) = \frac{z}{z-1}$$

$$f(2) = \frac{2}{2-1} = 2$$

Evaluate $\int_C \frac{dz}{z^2 - 4}$

where (i) $C: |z+2| = 1$

(ii) $C: |z| = 1$

$$\boxed{[y=2, \text{ & } -2]} \text{ (i)}$$

$$\frac{dz}{z^2 - 4} = \int \frac{dz}{(z-2)(z+2)}$$

$$|z+2| = 1$$

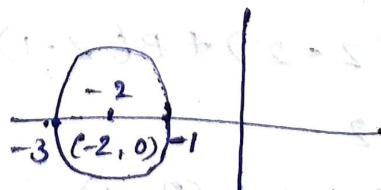
$$|x+iy+2| = 1$$

$$|x+2+iy| = 1$$

$$\therefore z = -2$$

is inside C

Centre $(-2,0)$



is outside

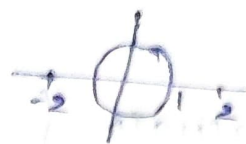
$$\int \frac{dz}{z-2} = 2\pi i f(2)$$

$$\int \frac{dz}{z+2} = 2\pi i f(-1/4)$$

$$= -\frac{\pi i}{2}$$

$$f(z) = \frac{1}{z-2}$$

$$f(-2) = \frac{1}{-2-2} = -\frac{1}{4}$$



$$|z|=1$$

since $z = -2, 2$ lies outside of $c: |z|=1$

$$\int \frac{dz}{(z-2)(z+2)} = 0$$

Cauchy's Residual Theorem

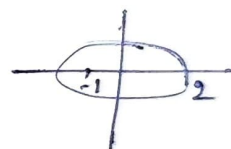
If $f(z)$ is analytic in a closed curve c except a finite no of singular points within c then

$$\oint_c f(z) dz = 2\pi i [Res f(a_1) + Res f(a_2) + \dots]$$

$$Res(m, a) = \frac{1}{(m-n)!} \lim_{z \rightarrow a} \frac{d^{m-n}}{dz^{m-n}} [(z-a)^m f(z)]$$

→ Evaluate $\int_c \frac{e^{2z}}{(z+1)^4} dz$ where c is the circle $|z|=2$ using Residual Theorem.

→ Since $z = -1$ lies in the circle $|z|=2$



$$Res(-1) = \frac{1}{(m-1)!} \lim_{z \rightarrow -1} \frac{d^{m-1}}{dz^{m-1}} [(z+1)^m \cdot f(z)]$$

$$Res(-1) = \frac{1}{(4-1)!} \lim_{z \rightarrow -1} \frac{d^{4-1}}{dz^{4-1}} [(z+1)^4 \cdot \frac{e^{2z}}{(z+1)^4}]$$

$$= \frac{1}{3!} \lim_{z \rightarrow -1} \frac{d^3}{dz^3} [e^{2z}]$$

$$= \frac{1}{6} \lim_{z \rightarrow -1} \frac{d}{dz} (4e^{2z})$$

$$= \frac{1}{6} \lim_{z \rightarrow -1} \frac{d}{dz} (4e^{2z})$$

$$= \frac{1}{6} \lim_{z \rightarrow -1} (8e^{2z})$$

$$= \frac{1}{6} (8e^{2(-1)}) = \frac{8}{6e^2} = \frac{4}{3e^2}$$

$$\int_C f(z) dz = 2\pi i [\text{Res}(-1)]$$

$$= 2\pi i \left[\frac{4}{3e^2} \right]$$

$$= \frac{8\pi i}{3e^2}$$

2) Evaluate $\int_C \frac{\sin^2 z}{(z - \pi/6)^3} dz$ where $C: |z| = 4$ using residue theorem

$\Rightarrow m=3 \quad a=\pi/6$

Since $z = \pi/6 = 0.523$ lies in the circle $|z| = 4$

$$\text{Res}(\pi/6) = \frac{1}{(m-1)!} \lim_{z \rightarrow \pi/6} \frac{d^{m-1}}{dz^{m-1}} \left[(z - \pi/6)^m \cdot f(z) \right]$$

$$\text{Res}(\pi/6) = \frac{1}{(3-1)!} \lim_{z \rightarrow \pi/6} \frac{d^{3-1}}{dz^{3-1}} \left[(z - \pi/6)^3 \cdot \frac{\sin^2 z}{(z - \pi/6)^3} \right]$$

$$= \frac{1}{2!} \lim_{z \rightarrow \pi/6} \frac{d^2}{dz^2} (\sin^2 z)$$

$$= \frac{1}{2} \lim_{z \rightarrow \pi/6} \frac{d}{dz} (2 \sin z \cos z)$$

$$= \frac{1}{2} \lim_{z \rightarrow \pi/6} \frac{d}{dz} (\sin 2z)$$

$$= \frac{1}{2} \lim_{z \rightarrow \pi/6} (2 \cos 2z)$$

$$= \frac{1}{2} (2 \cos 2\pi/6) = \cos \pi/3 = 1/2$$

$$\int_C f(z) dz = 2\pi i (\text{Res}(\pi/6))$$

$$= 2\pi i (1/2) = \pi i$$

3) Evaluate $\oint \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2 (z-2)} dz$ where C is the circle $|z| = 3$

$\Rightarrow z = 1, 2$ are lies in side circle $|z| = 3$

$$\text{Res}(m/a) = \frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} \left[(z-a)^m \cdot f(z) \right]$$

$a=1, m=2$